Changelog: v20 → v40

v21: survival search, free steps, routing, landing safety, rushes and the king's plan; v22: the judge's points budget, the queen, the centre rush; v23: per-map settings and the queen's survival; v24: portals; v25: dispersion; v26.0.0: a portal is a way out; v26.1.0: early portals; v26.2.0: splits of 2; v26.3.0: Leo's v27.2.0 merged; v27.0.0: eliminations; v27.1.0: the queen kill; v27.2.0: lanes, splits in dead ends, the queen's room; v31.0.0: Leo's v30 as the base, the queen first; v40.0.0: one bot from six

Summary

v21 was built in five rounds.

Part I, the survival search. A depth-first *survival search* (`survival.hpp`) asks, for each candidate move, whether our dragon can keep moving for its own length plus two more turns, playing out the engine's movement rules exactly. Its result feeds the safety filter (`critical_ok`) and the position score. Two supporting changes came out of testing: traps are priced by whether a split-escape is actually possible, and workers of length ≥ 8 may use the reserved escape slot at the unit cap.

Part II, the free-steps rule and routing.

- The game update's sprint rule ($\lceil L/4 \rceil$ free steps a turn) is now modelled everywhere the bot reasons about movement: its own moves, enemy reach and the simulator.
- The bot uses the rule itself: besides single steps it weighs *free moves* of several steps.
- Pearl routing (`route.hpp`) is rebuilt around four ideas:
  - time-aware paths: bodies are passable once they have moved on;
  - arrival measured in turns rather than steps;
  - who gets to each pearl first, by real paths from every visible head;
  - a survival search from where each route ends, so that a route that would box us in, above all the king, is never chosen.
- Testing showed free moves could carry the king to the edge of its vision, next to enemies it could not see, so valuable dragons now price that exposure.

Part III, landing safety and teammate memory. Free moves never leave the 7×7 window, but they can *end* at its edge. Long snakes then got boxed in, mostly by allies they could not see, 5–10× more often than after a single step. The changes:

- bodies are remembered for as long as they will really be there;
- allies' moves are modelled from what we see and from their status packets;
- long snakes pay a price for landing next to tiles they cannot see;
- the king no longer outruns its feeders in the late game.

Part IV, going back for remembered pearls. A long king that left pearls behind, such as pearls 30 steps away around its own body, never went back: routes only looked 20 steps ahead, which is one turn for a king of 80. The changes:

- routes now reach a few turns of free steps;

- a *rush* follows the route for all our free steps, also over ground we cannot see now but have seen, priced by how likely it is that something has moved there since;
- bodies running out of sight are predicted where they continue.

This round also found and fixed a bug that predates v21. Dragon ids pass 4096 in long games and our 12-bit status packets wrapped them, so feeders lost track of the king and teams were wiped out.

Part V, the king's own moves. Three changes for the king and growers:

- the survival search now proves a way on for their whole length, not 20 moves;
- a dragon that does not know all of its body treats unseen tiles next to where it ends as possibly its own body;
- each of the king's best moves is weighed by a search over its own multi-step moves for the next two turns.

Measuring showed that kings rarely die to their own body alone. They get boxed in by other dragons and then split-escape about once in ten turns, often in chains (see §27).

The queens/win-condition update is **not** implemented.

v22 (Part VI), the judge's points budget. The judge meters every dragon's turn in CPU points (100M a turn); a turn over it gets no reply, and the dragon dies with "no valid action". Measured with the judge's own pricing, v21's long dragons went over on hundreds of turns a game (kings averaged 164M a turn, up to 1.2 billion). v22 reads its own spending off the judge's clock, which counts points, and stops each search at its phase's deadline. The same searches were also made cheaper: a survival-search position by 2.5×, the king's plan options by over 10×. No measured turn of v22 goes over 87M, and v22 keeps v21's lookahead wherever it fits: the budget changes the chosen move on 0% of turns below length 100 and about 8% at 150+.

File	Change
`survival.hpp`	**New**: the survival DFS (Part I); bodies-only timings for routing, free-step head reach, `survival_from` for route ends (Part II); remembered bodies' lifetimes, allies' known lengths and status packets (Part III); deep searches, unknown-body caution, the king's plan (Part V)
`route.hpp`	**Rewritten** (Part II): time-aware searches, ETAs from every visible head, turn-based route values, BFS-guided replay with a route-end survival check, free moves; no exploring free moves for the late-game king (Part III); route range in turns, rushes (Part IV); deep route-end checks for the king (Part V)
`main.cpp`	Movement-rule helpers; config; `Candidate` fields; turn order; safety filters; escape slot; `simulate_path`, threat maps, reach threats, sprints and the ally contest under the new rule; enemy-density tracking; body memory (`busy_until`, `remember_bodies`), `known_length`, ally density (Part III); fill rate and occupancy, remembered tiles in `simulate_path`, hidden segments, 12-bit ids (Part IV); deep-search and plan constants, `Candidate::plan_value`, plan hook (Part V)
`scoring.hpp`	`trap_cost`; new `position_value` branches; goal progress for moves of k steps; route progress; fight reach; `unseen_risk` (Part III: a box-in part for every dragon of 10+)
`explore.hpp`	Pearl sprints: survival search, new rule, contest by turns; `disperse_score` per step
`endgame.hpp`	Death-yield comment; "cheap way out" by paid steps; `feeding_score` per step; 12-bit ids (Part IV)
`coordination.hpp`	12-bit ids (Part IV)
`helper.hpp`	Replaced with the upgraded-protocol release (still `PROTOCOL_MAJOR = 3`). Against ours, only the `make_moves` comment differs: it now states the free-steps rule (the first $\lceil L/4 \rceil$ steps free, at least 2 segments kept), which the bot and `sim/engine.py` already follow. The compiled bot is byte-identical.
`sim/engine.py`	The free-steps rule (`FREE_MOVES` switch)
`sim/` probes	`probe_aggr.py` (metrics), new `probe_trapped.py`, `probe_bigdeath.py`, `probe_turn.py`, `scen_route.py`, `probe_freemove.py`, `probe_hitbody.py`, `ab_trap.py`, `probe_fill.py`; `scen_route.py` gains `loopback` and `ownpocket`; `probe_selfdeath.py`, `probe_kingtrap.py`, `probe_ksplit.py`, `probe_kroom.py`, `probe_kingself.py`, `probe_bodyk.py`

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Part I
The survival search

1 Movement rules the search is built on

These were confirmed before writing the search and are what `surv_dfs` simulates:

1. **Self-collision includes the tail**, and is checked *before* the tail advances. Within a multi-step move, each step is checked against the body as the previous step left it.
2. **Dragons act one at a time**, and each sees the actual board at its turn.
3. **Sprints** check collisions at every step and remove one extra tail segment per paid step (Part II: the first $\lceil L/4 \rceil$ steps of a turn are free).
4. **Pay before eat**: the validity check for a step happens before that step's pearl is eaten, so a pearl does not save a dragon on its death step.
5. **Death yield**: a dragon of length L always drops $\lceil L/2 \rceil$ pearls.
6. **No simultaneous entry**: two dragons never move into a tile at the same time.

2 New file: `survival.hpp`

Included inside `ReplayDerivedBot`, before `scoring.hpp`.

2.1 What it computes

```
struct SurvivalResult {
    int depth = -1; // most moves survived on any path (-1: not searched)
    int target = 0;
    bool complete = false; // every path searched (budget did not cut it short)
    bool ok() const { return depth >= 0 && depth >= target; }
    bool dead() const { return depth >= 0 && complete && depth * 4 < target * 3; }
};
```

- **Target depth**: $\min(L + \texttt{DFS_EXTRA_DEPTH},\ \texttt{DFS_MAX_DEPTH}) = \min(L + 2, 20)$, where L is our length after the candidate move. Surviving L moves means our whole current body has moved off its tiles; the +2 is margin.
- **ok**: some path reached the target.
- **dead**: the search finished in full and the best path is short of 3/4 of the target. This is a trap.
- Anything else (including a budget cut) is "unknown", and the old v20 heuristics decide.
- Our own moves are searched *one step a turn*. Free steps are optional, so we can always go that slowly, and that pace gives other bodies the most turns to move on.

2.2 State (new members)

- `other_free_at_`: per tile, how many moves the other dragons must have made before the tile is free (`INF`: never).
- `body_free_at_` (Part II): the same for *bodies only*, meaning other dragons' segments and remembered bodies. Our own segments and the head-reach margin are left out. Routing and the ETA searches use it.
- `surv_occ_`: our body during the search (occupancy counts).
- `surv_eaten_`: pearls eaten along the current path, so the same pearl is not eaten twice.
- `surv_nodes_`, `surv_turn_nodes_`: per-candidate and per-turn node counters.
- Part II: `surv_route_nodes_` (the route checks' own per-turn counter), plus `surv_counter_`, `surv_cap_` and `surv_offset_`, which select the counter, cap and time offset for the current search.

2.3 `prepare_survival()`: once per turn

Called from `play_turn`. Part II moved it before the objective map and routing (see §8.6). It resets both node counters and fills `other_free_at_` and `body_free_at_`:

1. A tile with a **remembered** (out-of-sight) body blocks for $\texttt{DFS_REMEMBERED_STEPS} = 3$ moves (both maps).
2. **Our own unplaceable segments** (`own_idx_ == -2`, never seen since birth) free after $\max(1,\ L - \texttt{known_body_})$ moves, since they are in the unknown tail (`other_free_at_` only).
3. **Every visible part of another dragon** is a wall (`INF`) by default (both maps).
4. For each **visible head**, the body is traced (`trace_body`). Segment i frees after $\max(\texttt{DFS_OTHER_MIN_WAIT},\ est - i + 1)$ moves, where *est* is that dragon's estimated length (both maps). Each dragon moves once between two of ours; the +1 covers it eating (its tail then stays); and nothing frees before our second move. Segments whose head we cannot see stay walls.

5. **Head reach**: tiles a visible head could be on before our next move get a value of at least 2, so they are not counted on for our first step. In Part I these were its non-reverse next tiles. In Part II they are every tile within `free_moves(est)` steps (`tiles_within`), since its body follows it there. This applies to `other_free_at_` only; where heads go after that is left to the fight search.

2.4 `survival(pm, heading)`: per candidate

- Builds our body after the move, head first: `pm.path` reversed, then `own_trail_`, truncated to `pm.length`. Any remaining length is an `unknown` count of tail segments we cannot place.
- Part II: the search itself is now `survival_from(body, unknown, heading, length, offset, route)`, which `survival` calls with offset 0 and the moves' budget.
- If the per-turn budget is already spent, it returns "not searched" (`depth = -1`).

2.5 `surv_dfs`: the search

- Stops with success at the target. It stops with `cut = true` when either budget runs out: `DFS_NODE_BUDGET` = 4000 per search, and `DFS_TURN_BUDGET` = 40000 per turn for moves. Route checks have their own `ROUTE_DFS_TURN_BUDGET` = 24000.
- **Children** are the (up to three) non-reverse directions whose destination is passable and known. The tile must also hold none of our body (tail included, per rule 1) and have `other_free_at_` $\leq$ *step* + `surv_offset_`. Kelp blocks, the map wraps, and known portals are followed. Unknown portals are not taken, since we cannot see where they lead.
- **Ordering**: children are sorted by *openness*, the count of free neighbours at the next step, most open first. In open water the first path reaches the target, so the search is cheap; the budget only bites in tight spots. This is a manual insertion sort over at most three elements (see §4).
- **Moving**: the head is pushed. If the tile has a pearl seen this round that is not already eaten on this path, we grow and the tail stays. Otherwise an unknown tail segment is consumed if any remain, else the real tail tile is popped.
- **Backtracking** undoes every change: the head push, the tail pop, the unknown count and the eaten marks.
- Records the deepest depth reached in `best`.

3 Part I changes outside `survival.hpp`

3.1 Configuration (`main.cpp`)

```
#ifndef LONG_WORKER_ESCAPE
#define LONG_WORKER_ESCAPE 8
#endif
constexpr int LONG_WORKER_ESCAPE_LENGTH = LONG_WORKER_ESCAPE;

// v21: survival search
constexpr int DFS_EXTRA_DEPTH = 2; // target = length + this ...
constexpr int DFS_MAX_DEPTH = 20; // ... capped here
constexpr int DFS_NODE_BUDGET = 4000; // per search
constexpr int DFS_TURN_BUDGET = 40000; // per turn (moves)
constexpr int DFS_REMEMBERED_STEPS = 3; // remembered body blocks this long
constexpr int DFS_OTHER_MIN_WAIT = 2; // no other segment frees before this
constexpr double P_DFS_SHORT = 0.9; // trap probability per unit shortfall
```

`LONG_WORKER_ESCAPE` sits after `ESCAPE_UNIT_RESERVE = 2`. Its comment records why it exists: boxed-in workers at the unit cap used to die whole, about 140 deaths of dragons of length 10+ per 200 rounds on a big map. Compiling with `-DLONG_WORKER_ESCAPE=1000` turns the feature off.

3.2 `Candidate` fields

```
int dfs_depth = -1, dfs_target = 0; bool dfs_complete = false;
bool dfs_ok() const { return dfs_depth >= 0 && dfs_depth >= dfs_target; }
bool dfs_dead() const { return dfs_depth >= 0 && dfs_complete
                            && dfs_depth * 4 < dfs_target * 3; }
```

3.3 Include

`##include "survival.hpp"` is added before `scoring.hpp`.

3.4 Escape slot policy (`escape_slot_available`)

Previously a worker always had to leave `ESCAPE_UNIT_RESERVE` = 2 unit slots free before it could split-escape. Now:

```
reserve = king ? 0
       : (valuable_now_ || length >= LONG_WORKER_ESCAPE_LENGTH) ? 1
       : ESCAPE_UNIT_RESERVE;
```

A worker of length ≥ 8 may use the second-to-last slot to split out of a trap, like a valuable dragon. Losing 8+ segments whole costs far more than one extra unit near the cap.

3.5 Safety filters

In `evaluate_single`, after the room facts are gathered, `survival(pm, direction)` runs and is stored in the candidate. The room part of `critical_ok` is now:

```
room_ok = dfs_ok()
      || (!dfs_dead() && exits > 0
          && space >= pm.length + unknown_body() + 2 && !trap);
```

- A move the search proves survivable is safe even when the flood-fill heuristics call it tight.
- A move the search proves a trap is unsafe even when they look fine.
- Unknown falls back to v20.
- The other conditions are unchanged: known destination, no unknown portal, no head trade, no occupied exit, and for valuable dragons no hard threats or blind portal.

The same rule, run from each sprint's end, decides `critical_ok` for escape sprints (with `hard_threats == 0`) and pearl sprints (with `!valuable || hard == 0`).

3.6 Scoring (`scoring.hpp`)

`trap_cost`. In v20 a trap was priced as if a split-escape were always available (2 segments plus a unit). At the unit cap it is not, and the dragon dies whole, so traps were underpriced by about 3×. Now:

```
bool const can_escape = L >= 4 && escape_slot_available()
                      && controller_.can_split(L - 2);
return can_escape ? seg(2.0 + Config::UNIT_VALUE_OURS) : death_cost();
```

`position_value`: new branches after the pocket-plan case.

- `dfs_ok()`: the room is proven. Only the `P_ONE_EXIT` risk is charged when `exits == 1`; the space and dead-end risks are skipped.
- Searched in full but short: `v -= risk(P_DFS_SHORT * shortfall, trap_cost())`, where $shortfall = (target - depth)/target$.
- Not searched, or cut short by the budget: the v20 exits/space/dead-end risks, unchanged.

3.7 Debug log and comments

- The `CAND` debug line ends with `dfs <depth> <target> C|-`, where `C` means the search was complete.
- `endgame.hpp`: "Full yield" was misleading, so the comment now reads: *Every death leaves* $\lceil L/2 \rceil$ *pearls on the body's tiles; dying on our own neck here (rather than in a trade) puts them where the king will eat them.*

4 How the search got here

1. **First version (optimistic).** Other dragons' tails freed at step 1 and other heads were ignored. Since `dfs_ok()` drops the `exits == 0` risk, dragons stepped onto tiles with no free neighbour.
2. **Fix: cautious timing.** Added `DFS_OTHER_MIN_WAIT = 2` and the $est - i + 1$ eating margin, and blocked other heads' next tiles for our first step.
3. **Fix: trap pricing.** Nearly all self-deaths were `TRAPPED_SELF` (no legal move) at ≥ 62 units, where workers cannot split-escape. `trap_cost` now checks `escape_slot_available() && can_split`.
4. **Escape slot.** Even correctly priced traps still happen at the cap, so long workers were allowed the reserve slot.
5. **Compiler warning.** `std::sort` on `std::array<Child,3>` triggered a GCC `-Warray-bounds` false positive. It was replaced with a manual insertion sort.

5 Part I results (old movement rule)

No-legal-move deaths in 200 rounds on seed 20 against the baseline:

Version	Trapped deaths	of which length 10+
v20	318	157
v21, cautious DFS	297	–
v21, + trap pricing	270	–
v21, + escape slot	238	92

- Walls around trapped heads were mostly allies (ally body 463–499, ally head 86–111), then own body (∼450), kelp (∼120) and enemies (∼25).
- Full games against v20: 2–2 without the escape slot; **4–2** with it, with accidental self-deaths about 27% lower on big maps.
- The scenario tests passed and the coil stress test failed exactly as v20 did; the maximum turn time was 5–13 ms.

Part II
The free-steps rule, free moves and routing

6 The free-steps rule

6.1 The rule as modelled

In one turn a dragon of length L may take $\lceil L/4 \rceil$ steps for free. Every step after those costs a segment (its tail loses one more) and is allowed only while the dragon is longer than 2. That gives 1 free step at lengths 1–4 (the old rule), 2 at 5–8, 3 at 9–12 and 5 at 20.

Assumption: the allowance is per turn, and L is the length at the *start* of the turn.

```
constexpr int free_moves(int length) { return std::max(1, (length + 3) / 4); }
constexpr int paid_steps(int length, int steps) { return std::max(0, steps - free_moves(
    length)); }
// The most steps it can take in one turn (eating nothing on the way).
constexpr int max_turn_steps(int length) { return free_moves(length) + std::max(0,
    length - 2); }
// Turns to take `steps` steps at full free speed.
constexpr int turns_for(int steps, int length); // ceil(steps / free_moves(length))
```

6.2 Where the bot now uses it

Place	Change
`simulate_path`	The pay check applies only to paid steps (`s >= free && length_now <= 2`). Length before step s is $L + eaten - \max(0, s - free)$; after the move, $L + eaten -$ `paid_steps`.
`play_turn`	The projected length sent in the status packet subtracts `paid_steps`.
Reach threats (valuable)	Hard reach is `max_turn_steps(lower_length)` (was `lower_length - 1`). Soft reach for a truncated body is `max_turn_steps` of the larger of the typical enemy length and that dragon's own estimate (what we see +2), still capped by `HIDDEN_BODY_REACH`.
`build_threat_maps`	An enemy head threatens every tile within its free steps (`tiles_within`); an ally's next step is still what crowds us. For a reach of 1 this is the same set as before.
Fight search	An enemy's sprint into our head reaches `clamp(max_turn_steps(len), 1, max(FIGHT_SPRINT_REACH, free_moves(len)))`.
Survival search	Head reach covers its free steps (Part I, `prepare_survival`).
Tactical (attack) sprints	Up to `min(max_turn_steps(L), MAX_ATTACK_STEPS)` steps; the tie-break cost is per *paid* step.
Escape sprints	Up to `min(max_turn_steps(L) - 1, MAX_ESCAPE_STEPS)` steps; the score is `(eaten - paid) * SEG`.
Pearl sprints	Only for pearls beyond our free steps (closer ones are free moves). Net is eaten minus paid. "Contested" means an enemy needs no more turns than we would at free speed (by ETA, §8.3).
Split escape	A "cheap way out" is any safe move paying at most 2 segments (was: at most 3 steps).
Simulator	`engine.py`: `FREE_MOVES = True` applies the rule (pay check and extra tail segment only past the free steps).

7 Free moves

7.1 Generation (`add_free_moves`, `route.hpp`)

Each turn, after the single steps and the tactical and pearl sprints (and before escape sprints), we weigh moves of 2 up to $\min(\texttt{free_moves}(L), \texttt{FREE_MOVE_MAX_STEPS} = 6)$ steps:

- A depth-first walk over step sequences (never reversing). Every step is through a tile we can see now, with no portal. Each prefix is checked with `simulate_path`, which uses the exact engine rules.
- One option is kept per (end tile, last direction), the best by a quick look (`quick_value`): food on the way, exits at the end, and goal progress (along the route by the pearls' own maps).
- The best $\texttt{FREE_MOVE_EVALS} = 6$ are evaluated in full by `evaluate_path`.

7.2 Evaluation (`evaluate_path`)

It gathers the same facts as `evaluate_single`: threats at the end, exits, time-aware space and region, known space (valuable), and the survival search. The move is priced as in `move_value`:

- goal progress for k steps (half strength in a fight);
- every pearl eaten on the way (`path_food`), minus paid steps (none);
- `position_value` at the end and, in a fight, `fight_value` from there;
- visits to the end tile, teammate spacing (unless it eats), and `BLOCKED_SPAWN` for every path tile about to spawn;
- the enemy-distance pull.

`critical_ok` and `safe` follow the single-step rules. The new `Candidate::free_move` flag makes the chosen move log as `FREE_MOVE`.

7.3 Goal progress for k steps (`scoring.hpp`)

`goal_progress` now takes the `PostMove` instead of a `Step`, so the same code prices one step or several. With $k = 1$ every term is as before.

Goal	Progress
EAT	Steps still to go along the route, before minus after, clamped to $\pm k$ (§8.5); `ROUTE_TIE` if the move follows the checked route path.
HUNT	Grid distance gained to the target (unclamped, as before).
FEED	`feeding_score(dest, k)`: the gain clamped to $\pm(k+1)$ (was ± 2).
EXPLORE	Objective-map gain capped at $k+1$ (was 2); `disperse_score(dest, k)` clamped to $\pm k$; rendezvous unchanged.

8 Routing (`route.hpp`)

8.1 Time-aware paths

`timed_bfs(pm, limit, target, out)` gives the fewest steps for our head to reach a tile from where a move `pm` leaves us (or from where we are now). Every body is placed where it will be by then:

- Our own segment j (after the move) is passed once the tail has moved past it: at least $\mathit{length} - j + 1$ more steps.
- Another dragon's segment is passed once that dragon has made `body_free_at_` moves, one a turn. Our step is assumed to come in turn `turns_for`(*step*), at full free speed, plus the current turn if a move is already being made. Going that fast is pessimistic about other bodies.

- Kelp, tiles we know nothing of, and marked traps block.

It is used in three places:

- `route_head_dist_`: the map from our head, out to 30 steps (`prepare_etas`).
- The pearls' own maps (legs between pearls): a body tile is passable if our time-aware map reaches it, since it will have moved on by then. The first leg of every route uses `route_head_dist_` directly.
- Route progress for candidate moves (§8.5).

8.2 Arrival in turns

A pearl reached at step t is reached in `turns_for`(t, L) turns. The discount (0.92 for keepers, 0.85 for workers) is now *per turn*; for $L \leq 4$ this is the old per-step rule. Orders that arrive in the same turns are told apart by a 10^{-4}-per-step tie-break. Candidate pearls are filtered by time-aware steps within the route radius (20 for keepers, 14 for workers).

8.3 Who gets there first

`prepare_etas()` runs once per turn, before the objective map. It computes our time-aware map and one search per visible head (`head_eta_bfs`, 30 steps):

- That head's own segments pass once it has moved on.
- Our segments pass once we have made that many moves (we move before each of its turns).
- Others pass per `body_free_at_`.

Enemies are merged into `enemy_eta_turn_` and `enemy_eta_steps_`; allies are kept per head (`ally_etas_`).

The results are used in four places:

- **Route values**: a pearl an enemy reaches in fewer turns than our arrival on that route is worth `ROUTE_CONTEST_LOST` $= 0.4$ of itself. Before, any pearl closer to an enemy by *grid* distance than to us by path was halved. We win ties: every other dragon's t-th turn comes after ours.
- **Keepers**: the fight-risk factor (0.4) now needs an enemy within `ROUTE_ENEMY_NEAR` $= 4$ *path* steps of the pearl.
- `pearl_taken_by_ally`: turns at each dragon's own free speed, by time-aware paths for us and visible allies. Allies out of sight still use grid distance, now in turns. Ties still go to the lower id, a rule both dragons follow.
- **Pearl sprints**: contested by turns (see §6).

8.4 Replay and the route-end check

`replay_route` walks a route as the engine would move us, with our body trailing and growing:

- Each leg follows a shortest time-aware path from wherever the replay has got to (`replay_leg`): a BFS that knows the replayed body's segment indices, with the most open first step preferred.
- If the growing body gets in the way, the leg is searched again, at most 3 times per leg.
- A replay may take at most `ROUTE_REPLAY_SLACK` $= 8$ steps more than the horizon.

From where the route ends, the survival search runs (`survival_from`) on the replayed body:

- The time offset is `turns_for`$(t) - 1$ other-dragon moves; the pearls the route ate are marked eaten.
- It uses its own per-turn budget, so routing never starves the move evaluation.

The route is accepted only if that search succeeds:

- **King and growers**: `ok()`, or cut short by the budget after reaching at least 3/4 of the target.
- **Workers**: not `dead()`.
- **Budget already spent**: the old check (at least one free neighbour) decides.

The fallback (the nearest pearl on its own) goes through the same check. The replayed tiles are kept in `route_path_`, its first tile in `route_next_cell_`, and the order in `route_seq_`.

8.5 Route progress

`route_remaining(pm, limit)` counts the steps still to go after a move:

- time-aware steps (`timed_bfs`) from where the move leaves us to the first route pearl it has not eaten;
- plus the legs between the rest, skipping pearls eaten on the way.

`route_remaining_now_` is the same from here. EAT progress is the difference, clamped to $\pm k$; a move that eats the first pearl and carries on toward the second is credited for both. `follows_route(pm)` gives `ROUTE_TIE` when the move's tiles are a prefix of `route_path_`. The old `route_dist()` is removed.

8.6 Turn order

```
update_explore_target();
prepare_survival(); // v21: when other dragons' bodies move on (survival.hpp)
prepare_etas(); // v21: who reaches which tile first (route.hpp)
build_objective_map();
plan_route();
choose_goal();
...
add_tactical_sprints(candidates);
add_pearl_sprints(candidates);
add_free_moves(candidates); // v21
add_escape_sprints(candidates);
```

The debug `ROUTE` line now shows `rest <route_remaining_now_>` and `end_dfs <depth>`.

9 Valuable dragons: what they cannot see

A test game showed the danger. A king of length 111 made a 6-step free move to the edge of its window and landed next to an enemy of length 6 it could not see; the enemy rammed it with a free 2-step move. A single step never ends closer than 2 tiles to the window's edge; a free move can end on it.

`unseen_risk(pm)` (`position_value`, valuable dragons only) prices this.

- Count the tiles within `UNSEEN_RADIUS` = 3 steps of where we end that we cannot see now. Every single step has exactly one; an end at the window's edge has up to a dozen.
- Each tile is a chance of `P_UNSEEN_ATTACK` = 0.5 times the enemy heads we see per tile. That density is a running average (`enemy_density_x1e4_`, updated in `observe_heads`).
- The total is capped at `P_UNSEEN_CAP` = 0.3, times `death_cost()`.
- A blind portal exit is not counted here, since portals are priced separately.

A second king death (length 85) came from a body running out of sight. It was assumed only as long as a typical enemy, reaching 3 steps, but it was 8 long and paid 3 segments to reach 5. The soft reach now also uses that dragon's own estimated length (see §6).

10 New and changed constants (Part II)

```
constexpr int FREE_MOVE_MAX_STEPS = 6, FREE_MOVE_EVALS = 6;
constexpr int UNSEEN_RADIUS = 3;
constexpr double P_UNSEEN_ATTACK = 0.5, P_UNSEEN_CAP = 0.3;
constexpr double ROUTE_CONTEST_LOST = 0.4;
constexpr int ETA_RADIUS = 30;
constexpr int ROUTE_DFS_TURN_BUDGET = 24000;
constexpr int ROUTE_REPLAY_SLACK = 8;
```

`main.cpp` also gains `##include <tuple>`, the four rule helpers, `tiles_within` and `enemy_density_x1e4_`. Its header comment adds: *v21 (continued): the ceil(L/4) free-steps rule throughout (free_ moves); moves of several free steps; routing with time-aware paths, arrival in turns, who gets there first, and a survival check where each route ends (route.hpp).*

11 Part II testing (new movement rule, against v21 Part I)

11.1 Routing scenarios (`sim/scen_route.py`)

`room`. A king of length 15 next to a fully visible 3×3 walled room with a one-tile door and 3 pearls inside (eating them leaves 18 segments in 9 tiles), plus 2 pearls in the open.

- Both versions eat the open pearls first.
- The Part I version then plans a route into the room and spends 4+ turns walking at it; its move-level safety stops it at the door.
- The new version rejects every route into the room and goes exploring elsewhere.

`contest`. A worker of length 6 (2 free steps) and an enemy head 2 tiles from pearl P1 by grid distance, but behind a kelp wall (30+ steps by path); pearl P2 is 1 step further for us and uncontested.

- The Part I version gives P1 up to the enemy and eats nothing in 4 turns.
- The new version takes P1 and then heads for P2.

11.2 Games against v21 Part I (both under the new rule)

Seed	Map	New as	New max	Old max	Result
21	20×36	A	177	90	new wins
21	20×36	B	174	98	new wins
22	18×25	A	–	–	new wins (old eliminated, round 86)
22	18×25	B	–	–	new wins (old eliminated, round 76)
20	56×53	A	296	97	new wins
20	56×53	B	293	142	new wins
30	44×61	A	220	89	new wins
30	44×61	B	259	100	new wins

8–0. These games were played before the one-line route-end offset fix (§11.5). Seed 30 re-run with the final build: new wins both again (max 290 vs 95 as A, 310 vs 136 as B). By round 500 the new build is down to 1–2 units (its workers have fed the king), against the old build's 9–11. On total length the old build was ahead in both games (638 vs 562, 836 vs 310), so under the current tiebreak (longest dragon first) the new build wins, but the totals are worth watching once the queen rule is implemented.

11.3 Metrics (seed 20, 56×53, 200 rounds)

	New	v21 Part I
Pearls eaten	8791	4611
Total length at round 200	1052	627
Segments lost to enemy trades	2382	1044
Our dragons of 16+ hit by enemy-started trades	46	9
Accidental self-deaths (segments)	294 (3896)	251 (2012)
Trapped deaths, length 10+	122	68

Free moves almost double the food. They also make our workers far longer at the unit cap: 40–140 where the old build's are under 50. Long workers are juicy targets and die whole when trapped. Over full games, self-deaths of dragons of 16+ ran about 2.5× the old build's (137–169 vs 62–67 per game on big maps).

11.4 Other checks

- The original scenario tests (`scen.py`) all give the same outcomes as before.
- Coil stress test (`stress_box.py`, 30 trials): no king self-deaths, for either build.
- Turn time (seed 20, 250 rounds, debug build): at most 6.1 ms, 1.4 ms on average (the Part I build: 2.7 ms, 0.9 ms).
- No crashes, timeouts or pay deaths in any game.

11.5 How Part II got here

1. The first version let the king make long free moves to the window's edge. The two king deaths above led to `unseen_risk` and the estimate-based soft reach.
2. `std::unordered_map<int, std::pair<int, Direction»::operator[]` does not compile (`Direction` has no default constructor), so `insert_or_assign` and `at` are used.
3. The route-end search first counted the other dragons' moves one turn ahead (offset `turns_for`(t) instead of `turns_for`$(t) - 1$). That was slightly optimistic and is now fixed.

Part III
Landing safety and teammate memory

12 What was actually happening

The concern was that long snakes, able to take 20 free steps, move far past their window and so move blindly into teammates. They do not.

- Free moves are capped at `FREE_MOVE_MAX_STEPS` = 6 steps.
- Every step must be on a tile visible this turn, and `simulate_path` checks it against the board as it is.
- Dragons act one at a time, so nothing moves while we move.

What *is* blind is the landing. A multi-step move can end at the window's edge. Before our next turn, whatever is just outside the window (heads we cannot see, bodies we have forgotten) can close in, and the survival search cannot see it either. Bodies were also forgotten after 2 rounds, however long they were.

Measured (`probe_freemove.py`, seed 20, 200 rounds): how often a dragon was boxed in (no legal move) on the turn right after a move the survival search had called safe.

Length	After a multi-step move	After a single step	Multi-step, after Part III
10–29	28 / 2379 = 1.18%	1 / 959 = 0.10%	14 / 2536 = 0.55%
30+	4 / 803 = 0.50%	1 / 807 = 0.12%	4 / 897 = 0.45%
under 10	6 / 1673 = 0.36%	18 / 4561 = 0.39%	10 / 1592 = 0.63%

The walls around those heads were allies' bodies on 569 sides, our own body on 451, kelp on 123, ally heads on 81 and enemies on 44. Direct collisions with allies were rare (about 5 per 200 rounds).

13 Body memory

- `CellMemory::busy_until`: the round from which a body seen on the tile should be gone. It is set to $now + \texttt{OCCUPIED_MEMORY} + 1$ when the body is seen, which is the old 2-round memory.
- `remember_bodies()`, after `observe_heads`, refines it. Segment i of a visible head's body lasts $\lceil (len - i)/pace \rceil + 1$ rounds, at most $\texttt{BODY_MEMORY_MAX} = 30$, where:
  - *len* is `known_length` (for an ally, its last status report within 10 rounds, if longer);
  - *pace* is half the dragon's free steps for an ally (ours take several steps on most turns), and 1 for anyone else.
- A visible segment of an ally whose head we cannot see, but whose length we know, lasts half that length at its pace, clamped to 3–$\texttt{UNPLACED_BODY_MEMORY_MAX} = 8$ rounds.
- `body_remembered(m)` and `body_rounds_left(c)` replace the old fixed 2-round test. They are used by `remembered_occupied`, `blocks_pearl_path`, the time-aware space search and blind portal exits.
- The survival search frees a remembered tile after its rounds left (was a fixed 3); routing's `body_free_at_` does the same.
- The crowd count around route pearls still uses recent sightings only.

A first version assumed every body moves one step a turn. Measured over three maps, that was *worse* than having no long memory: our allies move faster, so it kept tiles closed too long and made the survival search pessimistic. The paced version measured slightly better than no memory.

14 Where allies can be before our next move

In `prepare_survival`, the step-1 head reach now uses what we know:

- **Allies**: min(`free_moves(known_length)`, `FREE_MOVE_MAX_STEPS`), since our allies never move further. **Enemies**: `free_moves(est)`.
- **Traced bodies** of visible heads use `known_length`.
- **Allies out of sight** that sent a status packet this round or the last: every tile within their reach of the reported head is blocked for our first step. The reach is doubled when the packet is from last round and the ally moves before us, since it has moved once since. The reported tile itself holds its body for $\min(30, len - 1)$ moves.
- `tiles_within` gained `any_first` (direction unknown) and now uses a set.

15 Landing next to tiles we cannot see

`unseen_risk` is now charged for every dragon from `position_value`, in two parts. With n the tiles within `UNSEEN_RADIUS` $= 3$ steps of the end that we cannot see now:

- **Ram** (valuable dragons, as in Part II): $\min(0.3,\ n \cdot$ `P_UNSEEN_ATTACK` $\cdot$ *enemy density*$) \times$ `death_cost()`.
- **Box-in** (any dragon of `UNSEEN_BOX_MIN_LENGTH` $= 10+$): $\min(0.3,\ n \cdot$ `P_UNSEEN_BOX` $\cdot$ (*ally* + *enemy density*)$) \times$ `trap_cost()`, with `P_UNSEEN_BOX` $= 0.05$. `ally_density_x1e4_` is a running average like the enemy one.

Every single step has $n = 1$, so single steps are not affected relative to each other.

Tried and dropped: a hard rule that long dragons end a free move at most 2 tiles from where they started. It measured no better than the price above.

16 The late-game king keeps to its feeders' pace

From `GATHER_ROUND` (380), the king and growers take no free moves while their goal is EXPLORE (outside a fight). They still take them to eat along a route, to fight, and to escape. Without this, the `feeder_near` scenario failed: a king of length 15 roamed 3–4 steps a turn and its length-4 feeder, at 1 step a turn, never reached it.

17 Part III results

Six large maps, 200 rounds each, both builds against the Part I build (`ab_trap.py`, seeds 20, 30, 37, 23, 27, 50):

	Part II build	Part III build	Change
Trapped deaths, length 10+	581	497	−14%
Accidental self-death segments	18054	16669	−8%
Total length at round 200	5237	5334	+2%
Segments lost to enemy rams	14320	15664	+9%

More long snakes stay alive, and some of them are then rammed instead.

Head-to-head against the Part II build, full games on large maps:

- the first version of these changes: 2–2;
- the paced memory: 1–3, every game by a small margin;
- the final build, seed 20 both sides: 1–1.

This is about even: the gain shows up in long-snake survival, not yet in wins.

Also found: every "ran into another body" death of ours is a single step through a portal whose exit we cannot see, onto an ally's body (`probe_hitbody.py`). About 40% of blind crossings die, the same before and after this round. This risk predates v21 and is priced at `P_BLIND_PORTAL` $= 0.3$.

Checks: the scenario tests give the same outcomes as before, including `feeder_near`, where the king is fed again. The cleaned-up build was confirmed move-for-move identical to the tested one before the late-game change.

Part IV
Going back for remembered pearls

18 What was wrong

In a game, a long king knew of pearls behind it but never went back for them.

- **Range in steps.** Routes considered only pearls within `KEEPER_ROUTE_RADIUS` = 20 steps (whole route: 28). A king of length 80 takes 20 free steps a turn, so a remembered pearl 30–40 steps away around its own body was never a route. The king fell back to exploring, and nearby unexplored ground (cost 12) always looked closer than a pearl 30+ steps away.
- **Speed capped by the window.** Even with a route, a free move only crosses tiles visible this turn. Towards a target outside the window that is 3 tiles of progress a turn, whatever the free steps.

Test (`scen_route.py loopback`): a king of length 40 is made to walk 60 steps west past three pearls, leaving them 22–25 steps behind with its body between, and then plays on its own.

Build	What happens
Before	No route: it explores away south and never eats them (20 turns watched).
Route range only	Goes back at 3 tiles a turn and eats all three 8 turns after taking over.
Final	Goes back and rushes 8 steps at once when the ground is quiet: all three eaten in 6–7 turns.

In this test the rush waits a few turns because the measured averages start from crowded-game values (see below).

19 Route range in turns

`route_radius()` and `route_horizon()` replace the fixed values: the old ones, or this many turns of our free steps, whichever is further:

- keepers: 4 turns (radius) and 6 turns (horizon);
- workers: 3 and 4 turns;
- capped at `ROUTE_MAX_RADIUS` = 100 and `ROUTE_MAX_HORIZON` = 140 steps.

A king of 40 now plans 40 steps ahead (60 for the whole route); one of 80, 80 and 120. Our time-aware map (`route_head_dist_`) is searched out to the same radius. Dragons with up to 5 free steps are unchanged.

20 Rush

`add_route_rush` runs when the goal is EAT. It follows the checked route path (`route_path_`) for all our free steps, never through a portal.

- **Moves weighed**: all of it, stopping on each route pearl, and the part we can see. A fully visible move of up to 6 steps is skipped, since it is already among the free moves.
- **Remembered tiles**: with `remembered_ok`, `simulate_path` also accepts a step onto a tile we cannot see now, if we have seen it, know the edge into it, and remember no body on it. Those tiles are returned as *blind*, and a pearl remembered on one counts as eaten (its tail staying put is the cautious case for self-collision).

- **Scoring**: `evaluate_path`, plus remembered pearls on blind tiles at their freshness, minus the chance of hitting something times `death_cost()`.
- **Safety**: a rush with more than `RUSH_MAX_RISK` = 8% (valuable dragons: 2%) chance of a hit is not safe. Rushes are exempt from the known-space penalty, since they price unseen tiles themselves.
- The chosen move logs as `RUSH`.

Chance a blind tile is taken: a tile last seen k rounds ago risks

$$\min(k \cdot \textit{fill rate},\ \textit{occupancy},\ \texttt{P_BLIND_TILE_CAP} = 0.5),$$

and a tile our own body is on now risks 0. Nobody else can be there, and nothing else moves while we do, so following our own body round is safe however far it goes. Both rates are measured in `remember_vision` as running averages:

- the fill rate: of the tiles we saw free last round and see again, the share now taken by another dragon;
- occupancy: the share of tiles in view holding another dragon.

They start at 3% and 20%. In a real game around the king they ran 3%/round and 23% in the midgame, and 1.5% and 15% late. So in crowds a tile seen 7+ rounds ago is a 20%+ risk and rushes are rare; in quiet ground or along our own body they go.

21 Bodies running out of sight

`predict_hidden_segments()` runs after `remember_bodies`.

- A visible segment's facing points to the next segment towards its head. If that tile is out of sight, it holds the same dragon.
- If no visible segment follows a segment towards its tail, and only one tile out of sight could, that tile almost surely does.
- Both tiles are remembered as taken for as long as the segment next to them, so rushes, the survival search and routing treat them as bodies. A remembered body no longer needs the tile to have been seen.

A rush in testing hit an ally's segment 3 just outside the window, where its body ran out of sight. Rush collisions went from 5 to 2 in 300 rounds with this.

22 The 12-bit id bug (predates v21)

Status packets, beacons and king records carry dragon ids in 12 bits. Both teams share one id counter, and with free moves (more eating, more splitting) it passed 4800 in a game. The king, id 4821, was reported as 725. Feeders looked for a head with id 725, never found it, never fed it, and wandered at lengths 150–215 until they died in their own coils. The team was wiped out in 2 of 6 games.

The fix: `pid(id) = id & 4095` and `my_id()`. Every id we store (`SeenHead`, signals, the king record, the ally length cache) and every comparison between a stored id and a real one uses that form; comparisons between two real ids (a visible part against our own id) are unchanged. Two live dragons whose ids differ by exactly 4096 would be confused, which is rare.

23 Part IV results

Full games, large maps, both sides:

New build	Against	Result	Final max length (avg)
Range, rush, hidden bodies (no id fix)	Part III build	4–2, both losses team wipeouts (id bug)	–
Final (with the id fix)	Part III build	**4–0** (seeds 30, 37)	413 vs 245
Final	Id fix only	**5–1** (seeds 20, 30, 37)	348 vs 278

So both parts pay: the id fix keeps the king fed, and the longer routes and rushes add about a quarter to its final length on top.

Over six maps of 200 rounds (before kings matter) the variants differ by about ±3% in total length, within run-to-run noise. Rushes eat about 3% more pearls there.

Checks: the scenario tests are unchanged (`room`, `contest`, the feeder tests) and `loopback` passes. The cleaned-up build plays move-for-move like the tested one.

Part V
The king's own moves

24 Deep self-safety

Until now a move was "proven safe" when the survival search found $\min(L+2, 20)$ moves. For a king of 200 that proves almost nothing: its newest segments stay put for up to 200 moves, so it could walk into a ring of its own recent body holding 25 tiles and be called safe.

- `deep_dragon()`: the king (also before growers start at round 300) and growers.
- For them, `survival_from(..., deep)` searches to $\min(L + 2, \texttt{KING_DFS_MAX_DEPTH} = 300)$. The budgets are `KING_DFS_NODE_BUDGET` = 20000 per search and `KING_DFS_TURN_BUDGET` = 250000 per turn (`KING_ROUTE_DFS_TURN_BUDGET` = 80000 for route ends, which are deep for them too).
- In deep searches, another dragon's body that we cannot place (head out of sight) is assumed gone after `DEEP_UNPLACED_BODY` = 40 moves rather than never, or a long search would fail on it.
- `survival_from` now takes a budget pool (0 moves, 1 route ends, 2 plans) and a per-search node cap.

Test (`scen_route.py ownpocket`): a king of 60 walks (scripted, so it knows its body) east and then round a full ring; at hand-over its newest 23 segments enclose 25 tiles holding pearls, and its head is at the ring's one gap.

Build	Stepping into the ring
Part IV build	*safe* (20 of 20 moves proven); kept out only by its score.
Final	*unsafe*: only 23 of the 62 moves needed exist; heavy penalty.

Both builds wait outside and eat the pearls once the ring has moved on, with 3 pearls inside or 7.

25 A body we do not know

A dragon born long from a split (an heir) starts as a new process: no map memory, and of its body only what it can see. On average heirs knew 79% of their body (`probe_bodyk.py`). The rest is attached where the known body ends (`own_trail_.back()`). In deep searches, while some of it is still there, a tile we have never seen within $\min(\textit{unknown}, \texttt{UNKNOWN_BODY_RADIUS} = 10)$ steps of

that end counts as possibly our own body. (Multi-step moves never caused a body-model reset; `probe_bodyk.py` counted none.)

26 The king's plan

`plan_king_moves` runs after all move candidates exist, for deep dragons with 2+ free steps.

- For each of the `PLAN_CANDIDATES` = 8 best moves (safe ones first), the turns after it are searched. For turns 2..`PLAN_TURNS` = 3, a state's options are a shortest path to every tile within our free steps (at most `PLAN_MAX_STEPS` = 24, no portals). The search is time-aware against our own moving body. The `PLAN_BRANCH` = 4 best are kept, by food on the way and progress along the route (`plan_options`).
- Each option is played out exactly by `plan_extend`. Every step is checked against our body as the step before left it, tail included. A pearl, seen or remembered, eaten keeps the tail.
- Other bodies are timed cautiously (`plan_free_at_`): a tile another head can reach may hold its body for its length, not just for our next move. A plan should not count on gaps another dragon can fill.
- **A plan counts only if it is proven safe**: from where it ends, the deep survival search (budget pool 2, `PLAN_DFS_TURN_BUDGET` = 150000) must find a way on for our whole length.
- The best safe plan's *food* in its later turns, discounted `PLAN_DISCOUNT` = 0.85 a turn, is added to the move (`Candidate::plan_value`, ≥ 0; -1 means none was safe). Progress is left to the move's own score.

Two dead ends on the way

- **Counting progress in the plan made the king hold back.** Later turns "made" 10 steps of progress each through tiles the real move would price as risky, so a single step now left more for the plan to collect. In `loopback` the king crept one step a turn (pearls eaten at turn 78, not 66). Food only fixed it.
- **Cautious timing in every deep search** (not just in plans) made kings doubt every move in crowds. It lost 3 of 3 games against the Part IV build and was dropped. Used only inside plans it went 4–2; without it, plans went 1–5.

27 How kings actually get trapped

- **Self-deaths of long dragons by role** (`probe_selfdeath.py`): about 9 per game are kings or growers. Most are dragons born from a split one or two turns earlier, already boxed in.
- **Escape splits** (`probe_kingtrap.py`, `probe_ksplit.py`): the king escape-splits on about **1 in 10** of its turns (56–89 times a game, in both builds). Each split costs the head and 2 segments and restarts the king as a new process that knows little of its body.
- **Where they come from**: about two thirds are "blind" splits by young heirs that do not know their body. These are chains: the head alternates between the two ends of a body boxed in at both. The rest are kings that knew their body and were boxed in mostly by *other* dragons, often right after a multi-step move (`probe_kroom.py`).
- The deep check, the unknown-body caution and the plan did not change that rate.

28 Part V results

- **Final build vs the Part IV build**, six full games on large maps, both sides: **4–2**. Final king length averaged 346 vs 338, which is within the large game-to-game swings.

- Scenario tests unchanged; `loopback` still eats at turns 66–67; `ownpocket` as above.
- Coil stress test: no king self-deaths.
- Turn time: at most 36 ms (the king's plans and deep searches), 1.7 ms on average.

29 Known limitations and not yet done

- **Free-steps assumption**: per turn, from the length at the start of the turn. If the real engine counts differently, `free_moves` and `engine.py` are the two places to change.
- **Queens / win condition**: not implemented.
- **Teammate memory is partial**: we remember what we have seen and what status packets tell us (head position and length only). Where an unseen ally's body lies is guessed.
- **Blind portal crossings** die about 40% of the time in crowded games.
- **Rushes** use only free steps. Paying segments to rush to a remembered pearl is not considered: the pearl may be gone.
- The fill rate and occupancy start at crowded-game values, so a newborn long dragon does not rush for its first 30 turns.
- **The king's escape-split chains** (§27): about 1 split in 10 king turns, mostly by heirs that do not know their body. The most valuable next step for the king.
- **Crowding at the unit cap** is now the biggest leak. Workers can no longer split there and grow very long with free moves; they then die whole when trapped or rammed. This is the most valuable next step (for example, feeding long workers to the king or capping worker length).
- The fight search still gives the enemy single-step replies plus a sprint into our head; free repositioning of 2+ steps is not played out.
- Free moves avoid portals and stay inside our window.
- `unseen_risk` counts unseen tiles; it does not use what we saw there last turn.
- Route ends: the other dragons' future moves are unknown. A body whose head we cannot see stays a wall forever.
- Edges never seen are assumed open (as before), so a route into unseen ground can look better than it is until we see it.
- The survival search covers our own single steps only; other dragons are bodies that drain away plus a head reach.
- Pearls in the survival search count only if seen this round.
- Planned but not started: panic fallbacks.

Part VI
v21 → v22: the judge's points budget

30 What the judge measures

From the Timeouts page and the organisers' public repository (`unswcpmsoc/unswbc`, `pythoncli/unswbc/metering.py` and `sandbox.py`, which reproduce the judge):

- Bots run as WebAssembly (C++: `-O2 -msimd128`, libc++). Each instruction has a fixed price in *points*: most cost 1, loads and stores 2, division 3, calls 4–6, memory growth 50, bulk copies 10 plus 1 per 8 bytes.
- **100 million points per dragon per turn**, no carry-over. The turn runs from reading the round to the flush in `end_turn()`; work after that counts for the next turn. A process's first turn includes its own setup.
- A write to stdout costs 2.5M plus 4,000 a byte; a read 6 a byte. The helper buffers the whole turn into one write (one 2.5M charge a turn); `output_log` lines cost only their bytes.
- **The clock counts points**: `clock_time_get` returns points spent (1 ns = 1 point). So `std::chrono::steady_clock` inside the bot is an exact meter of its own spending.
- A turn over budget is stopped and delivers nothing. The action stays at its default, suicide, and the engine records `DragonDeathReason::NoValidAction`: the "no valid action" deaths.

The engine (`engine/src/actions.cc`) also uses `NoValidAction` for a paid step at length 2, a split that would leave fewer than 2 segments on either side, and a split at the unit cap. v21 already avoids all three: paid steps are capped at $L - 2$ (`max_turn_steps`), and every split goes through `can_split`, whose unit count is current at our turn. So the deaths were the points budget.

31 Measuring it

`unswbc run --sandbox` measures exactly, but the toolkit ships through PyPI, which this workspace cannot reach. So the judge's build was reproduced:

- wasi-libc, libc++ and compiler-rt built from source for `wasm32-wasi` with clang 18 (`wasm/setup_toolchain.sh`); the bot built with `-O2 -msimd128 -mbulk-memory` (`wasm/build_bot.sh`). Exceptions are off; the judge's clang 20 with wasm exceptions may cost a little more.
- The organisers' own `metering.py` adds the point counter (`wasm/meter/instrument.py`).
- `wasm/meter/runner.mjs` runs it under Node's WASI the judge's way: the clock returns points, writes and reads are charged, and each turn's points are logged.
- The bot is deterministic, so a game is recorded natively (each dragon's input, `rec_bot.sh`) and every dragon is replayed through the meter (`replay.py`, `analyze.py`). A `BOT_PROFILE` build adds the points of each phase (`prof_an.py`).

32 v21's turns

One recorded 500-round game (seed 20, 56×53): our dragons of length 30+ and a sample of shorter ones, 7,410 turns, in millions of points.

Length	Turns	Median	p90	p99	Max	Over 100M
2–9	1715	20.3	33.3	81.9	150.6	13
10–29	1700	28.6	53.0	95.5	116.3	8
30–59	2702	40.8	102.7	173.3	299.2	292
60–99	933	39.6	200.4	351.5	934.7	252
100–149	210	27.7	312.0	536.4	840.8	75
150–199	77	30.4	575.4	1217.5	1217.5	31
200+	73	285.6	578.8	850.8	850.8	61

Kings averaged **164M a turn** (max 1.2B). Of that:

- 84M (51%) was the king's plan, almost all in `plan_options`: for every tile within 24 steps it copied the path and looked for the tile along the whole body (up to 300 segments) and the route.
- 50M (31%) was routing: 14 whole-map searches for the pearls' fields and the route replays.
- A survival-search position cost about 2,800 points, mostly `step_from` (edge copies, portal lookups in a `std::map`).

Short dragons sat at 15–20M a turn, most of it map-wide setup.

33 The turn's clock and deadlines

- `clock_now()` is `steady_clock` in nanoseconds: points in the judge. Native builds (local tests) multiply by `NATIVE_POINTS_PER_NS` = 24, measured against the metered build (p10–p90: 17–32).
- `main` starts the turn's clock before reading the round (the first turn: at program start).
- Deadlines, counted from the turn's start:
  - the route's checks stop at `ROUTE_STOP` = 40M;
  - the moves' survival searches at `MOVES_STOP` = 66M (first 64M);
  - the king's plans at `PLAN_STOP` = 91M (first 82M);
  - anything else at `SEARCH_STOP` = 92M (first 86M).

  The results below were measured with the first values. §36 raised them to a 95M peak. The phases are absolute, so a turn whose setup is expensive just leaves less for the later phases.
- One survival search may spend at most a quarter of what its phase has left (at least `SEARCH_MIN_POINTS` = 2M). A search deep in a tight spot can no longer use up the phase before the escape sprints are searched.
- Where the clock is read:
  - `survival_from`: before starting, and every 128 positions;
  - `plan_king_moves`: before each candidate and each state's options;
  - `plan_route`: after each pearl's field (fewer pearls are kept when time is short) and before each replay.
- **A search cut by time is treated like one cut by its node budget**:
  - a candidate's survival falls back to the room estimates;
  - a route end falls back to the free-neighbour check;
  - a plan that cannot be proven adds nothing.

  The node budgets stay; the deadlines only make sure the turn is never over.

34 Cheaper searches

- **Neighbour table.** `step_from` for every tile and direction lives in `nbr_` (the tile stepped onto, `NBR_WALL`, or `NBR_BLIND` for a portal whose far end we do not know), plus `nbr_portal_`. It is kept from turn to turn and refreshed once a turn, after the sonar and the vision are learned.
  - Only tiles whose edges changed are rebuilt, with their four neighbours: a step reads the far side's edge when its own is unknown.

  - A newly learned portal end rebuilds everything.
  - `step_from` itself reads the table, so every caller got faster. The survival search, the plan, the route fields, `timed_bfs`, `head_eta_bfs` and `replay_leg` index it directly.
  - `NBR_CHECK` compares it against the uncached `step_from` after the refresh and again mid-turn: no mismatch in two games. A deliberately broken refresh gives 133.

- **`plan_options` in one pass.**
  - Our body's tiles go in a stamped table (O(1) per tile instead of a walk along the path).
  - Food and the route pearls eaten (a bitmask) are carried along the search tree.
  - The route's remaining length is cached per set of pearls eaten: only its first leg depends on the tile.
  - Only the best `PLAN_BRANCH` options are kept, without sorting all of them, and nothing is allocated per call.
  - One behaviour change: pearls on tiles the plan already ate no longer count as food again.

- **Hash maps out of hot loops**: `replay_leg`'s parents and `head_eta_bfs`'s segment indices are stamped arrays, and `route_bfs` uses a vector queue.

- **The survival search's body is a ring buffer** instead of a `std::deque`. `SURV_CHECK` runs both on every search: no mismatch; a broken `back()` gives 22. It saved only about 4% a position.

A survival-search position now costs about 1,100 points (was 2,800). For kings of length 60+, `plan_options` fell from 103M to 5–12M a turn and the route fields from 43M to 8–12M.

35 Part VI results

Points (the same game and dragons as above, release build):

Length	Turns	Median	p90	p99	Max	Over 100M
2–9	1715	14.4	18.5	31.4	47.5	0
10–29	1700	17.8	24.1	31.1	36.6	0
30–59	2702	21.5	33.4	43.0	83.5	0
60–99	933	22.0	44.9	64.7	85.4	0
100–149	210	21.3	60.6	85.5	85.8	0
150–199	77	23.1	85.3	86.0	86.0	0
200+	73	59.4	85.2	85.6	85.6	0

- Kings now average 36.5M a turn: route 10.5M, plan 9.7M, free moves 4.5M, objective map 4.4M.
- Over four recorded games (15,225 turns of the dragons that reached length 40), the worst turn is **86.6M**. With the earlier 82M stop, every turn of every dragon in two v22 games (52,359 turns) stayed at or under 82.7M.
- Peak memory is 2.1 MB of the 48 MB allowed.
- The work after the last deadline (choosing, the output) is under 1M. The 13M margin is for the judge's build, which is not exactly ours.

How much lookahead the budget costs. On recorded inputs, v22 with deadlines off (`BOT_UNLIMITED`) chooses the same move as v21 on 2,244 of 2,258 turns. The 14 differences are all in one 200+ king and come from the plan's new tie-breaks. With deadlines on, v22's move differs from the unlimited one on:

- 0 of 2,047 turns at length < 100;
- 1 of 88 at 100–149;
- 7 of 83 at 150+.

At length 200+, the plan is cut on 8 of 73 turns and a single search's share on 14. The route and the moves are rarely cut (3 and 1).

Games. `engine.py --tle-ms` kills a dragon whose turn takes longer than that many wall-clock milliseconds, as the judge would. 4.5 ms is about 100M points at 24 points/ns. Each game ran on its own (`ab_tle.sh`).

Games	Result	Longest at the end (v22 / other)	Limit deaths per game
v22 vs v21, limit on: seeds 20, 6, 27, 30, 20 (v22 second in the last two)	**5–0**	81–139 / 0 (wiped out)	v22: 1–2. v21: 105–278, of them kings 49–99
v22 vs v22, limit on: 2 games	–	–	0
v22 vs v21, *no* limit (v21 unmetered): 3 games	0–3	170–194 / 392–456	–
v22 with deadlines off vs v21, no limit: 2 games	1–1	251, 354 / 271, 299	–

Two more limited games, recorded for the replays above (seeds 6 and 27), also went to v22 with v21 wiped out. **Correction (see §41):** native v22 in these simulator games ran without its searches, because of a bug in its native turn clock. So the "no limit" rows compare v21 with a v22 that had no lookahead, and do not measure what the budget costs. The metered replays, the move comparisons (run from recorded files) and your sandbox runs were not affected.

Under the limit, v21 is wiped out in every game. Its long dragons and kings die to the meter, which is what was seen in real matches. Without a limit, v21's unmetered lookahead (up to 1.2B points on a long king's turn) does buy strength. v22 gets as much of it as 100M allows.

v22's own limit deaths (7 in the 5 games against v21, none in self-play) took 4.9–5.8 ms, twice about 30 ms, and one was not timed. They are wall-clock noise of the local machine: native v22 stops its searches at about 3.6 ms, and the metered replays show no turn over 86.6M.

36 Checked in the sandbox, and a 95M peak

The official sandbox (`unswbc run -sandbox`, v21 as team A, v22 as team B):

Team	p50	p99	Mean	Max
v21	26.3M	52.1M	27.0M	99.7M
v22	17.1M	25.8M	17.1M	85.3M

- v22 won easily: its longest dragon reached 72, v21's 34.
- Seven v21 dragons died to the limit, all of them kings. A turn over budget shows as just under 100M because the sandbox stops it there.
- v22's 85.3M peak agrees with the metered replays here (86.6M), so the rebuilt toolchain prices the bot like the judge.

The peak is the plan's deadline plus a fixed tail. The metered replays were re-paired line by line: on replays from a file, the runner's round labels can run ahead, because the bot's input buffer reads ahead.

- The plan overshoots its stop by at most 0.8M (it ended at 82.81M at most against an 82M stop).
- Everything after it costs at most 4.1M on the 73 turns whose plan ended past 75M. That is the choosing, `send_status`, and the 2.5M write plus 4,000 a byte. The profiling build's own line is included; the release build is about 0.7M less.

So `PLAN_STOP` = 91M with `SEARCH_STOP` = 92M. `SEARCH_STOP` is kept just above the plan's stop, because a search after the plan (a split escape, say) could otherwise run to it and then pay the tail.

Measured over the same four recorded games (15,225 turns of the dragons that reached length 40):

- p50 18.2M, p99 66.8M, max **95.05M**;
- one turn over 95M, none over 96M.

This leaves 5M for anything the judge prices differently from the rebuilt toolchain. Its own run measured 85.3M where the replays gave 86.6M.

37 Tools added

- `wasm/setup_toolchain.sh`: builds the wasm32 sysroot (wasi-libc, libc++, compiler-rt builtins) with the system clang.
- `wasm/build_bot.sh SRC OUT.wasm [flags]`: the judge-like build.
- `wasm/meter/instrument.py` (with the organisers' `metering.py`): adds the point counter.
- `wasm/meter/runner.mjs`: the metered runner (`METER_BUDGET`, `METER_LOG`).
- `wasm/rec_bot.sh`, `wasm/replay.py`, `wasm/analyze.py`, `wasm/prof_an.py`: record, replay and summarise.
- `sim/engine.py --tle-ms`; `sim/ab_tle.sh`.
- Build flags:
  - `BOT_PROFILE`: a `PROF` line of points per phase, sub-timers, nodes and the deadlines hit;
  - `NBR_CHECK` and `SURV_CHECK`: the equivalence checks above;
  - `BOT_UNLIMITED`: no deadlines, for tests only.

With the official toolkit installed, `unswbc run <map> a b --sandbox` gives the judge's own numbers. It is worth running once on a long game to confirm the margin.

38 v22: known limitations and next steps

- **Setup costs every dragon 12–15M a turn**: objective map, survival preparation, ETAs and parsing. It is unguarded but bounded by the map size; trimming it would give every phase more.
- Long kings (200+) still have their plan cut on about one turn in nine (8 of 73). The next savings are free moves (about 15M on those turns), the route fields (12M) and the plan's option search.
- The deadlines are fixed shares of one turn. Search depth could be traded between phases by role. The budget resets every turn; nothing can be saved for a hard turn.
- The native factor (24 points/ns) is a median. Native test games only approximate the judge; the metered replay is the reference.

Part VII
v22 (continued): the queen

39 The rule

From the engine (`engine/src/scoring.cc`), at the last round the ranking is:

1. the queen's length (dragon 0 or 1, the first dragon each team's map lists; a dead queen counts 0);
2. the longest dragon;
3. total length.

A team that is wiped out loses at once. A split keeps the parent's id with the head part, so a queen that never splits stays the queen. The official maps give each team 1 to 9 starting dragons (`arena` and `Colosseum`: 1; `default`: 4; `australia`: 9).

40 What the queen does (`queen.hpp`)

- **Who**: `is_queen()` is `controller_.get_id() < 2`, the raw id. Our packets' 12-bit ids wrap past 4096, so heads we see carry their own `SeenHead::queen` flag, and beacons carry a queen bit (bit 18).
- **No splits**: no worker split (`should_split`), escape split (`play_split_escape`) or pocket raid (`pocket_plan_allowed`). Trap pricing assumes she cannot split out (`trap_cost`).
- **Growing the swarm first** (revised; see §44): until round `QUEEN_SEED_END_ROUND` = 75 she plays as a founder and makes the usual small worker splits. The head part, and her id, stays with her. (First version: only when she started alone, until we were 4 or round 60.)
- **The cycle, from round 75 until** `QUEEN_CYCLE_END_ROUND` = 400: a Hamiltonian cycle of a $w \times h$ block around her head (row 0, a serpentine over the other columns, back up column 0; h even).
  - The block is the largest size from 16 × 16 down, at most $\max(144,\ 2(L +$ `QUEEN_CYCLE_SLACK`$))$ tiles. Every step of its cycle must be an ordinary step (no kelp, no portal), by the neighbour table.
  - Of the placements, the one with the fewest other bodies on it and the fewest tiles never seen wins.
  - She walks it with as many of her free steps as are *queen-safe*: the usual safety, plus no enemy, ally, reach or soft threat at the end, and her whole length proven to have a way on (or two exits if that search was cut). Paid steps are never used.
  - The route search and the king's plan are skipped while she cycles.
- **Her body on the cycle**: walking it is safe only while every tile ahead is free of her own body by the time she reaches it (`queen_body_fits`). A segment i places back is gone after $L - i$ of her steps, one more for each pearl she eats on the way (the known pearls ahead plus `QUEEN_GROWTH_MARGIN` = 2); at least `QUEEN_CYCLE_MIN_FREE` = 4 spare tiles are required. She only follows the cycle while this holds. A new block (kelp found, or within 12 of her length) must pass the same test, and is never smaller than a cycle that still fits.
- **Off the cycle** (its next tiles are not queen-safe, or no cycle fits her): she takes the queen-safe move that ends nearest the cycle. If there is none, she takes the usual best move, as a deep-searching long dragon that never splits. She looks for a cycle that fits every `QUEEN_RETRY_ROUNDS` = 3 rounds.

- **Crown, from `QUEEN_CROWN_ROUND` = 380**: she stands for king, and she beats everyone in `offer_king` and `elected_king`, so beacons carry her and feeders gather on her. Before 380 she does not stand, and her length reports are ignored for the king/vice roles, so the team still has a roaming king. From 400 she retains as the king, everyone else feeds her, and her cycle ends. If she dies, the normal election takes over.

41 A bug in the native turn clock

The turn's clock started before reading the round. That is right in the judge: waiting costs no points there, and work after `end_turn` is charged to the next turn.

Natively, though, the clock ran on while the bot waited for every other dragon to move. So in every native simulator game, every turn began past all its deadlines and **v22 ran no searches**: no survival search, no plan, no route checks.

- This was found when the queen's candidates all showed an unsearched survival result.
- Native builds now start the clock once the round is read (`main`). The WebAssembly build, the one the judge runs, is unchanged.
- Affected: the native games in Part VI (the 5–0 under the limit, the 0–3 without one, `loopback` being slower).
- Not affected: the metered replays (recorded files and the points clock), the move comparisons from files, and your sandbox runs.

42 The simulator's pearls are far too rich

On the official maps, pearl beds and respawn gaps are set tile by tile (`TILE x y minGap maxGap`). On `default.map` most tiles respawn every 1–3,849 rounds (about 1,900 on average), and some every 1–255 or 1–385. The simulator's random maps use 4–60 on almost every tile: 30–100× more pearls. That is why simulator dragons reach 300–450 while the official sandbox game ended with a longest dragon of 72.

- `engine.py -pearl-scale k` (and `probe_queen.py`'s 6th argument) stretch the simulator's gaps by k.
- The simulator now ranks by queen length, then longest, then total, and reports `A_queen`/`B_queen`.
- Everything tuned in the simulator so far was tuned for the rich economy. Loading the official `.map` files into the simulator is the proper fix.

43 Queen results (simulator, so indicative only)

- With the simulator's rich pearls she grew to 35–77 by round 45–50, and every cycle filled up; she died off the cycle. This is what led to the body test above.
- With gaps stretched 30×, she walks the cycle on most turns (48 of her first 50 in one game). Her deaths there were enemy rams at length 3–7 (rounds 51, 107, 189): an enemy queen's 2-step sprint, and workers' attack sprints, on turns when no move was queen-safe.
- Queen build against an identical build without the queen (`NO_QUEEN`), 4 founders, 30× gaps: 1–1. Side A was wiped out in both games whichever build it was (a quirk of the stretched simulator to look into). Its queen survived in the game it won.
- Metered (the judge's pricing), over 250 queen turns: 11.2M points on average, 37.1M at most.

These need checking on official maps in the sandbox, with one founder (`arena`) and several (`default`).

44 The queen grows the swarm until round 75

A non-splitting queen hurt our early game: on maps with few founders, one of them stopped growing the swarm. Now:

- **Until round 75** she is an ordinary founder (`queen_seeding()` is just `is_queen()` and the round). She makes worker splits (she keeps her head, so she stays the queen) and joins the centre rush on small maps.
- She never makes pocket raids, and makes escape splits only when she is certainly dead (§45).
- **From round 75**: her Hamiltonian cycle. From 380 feeders gather on her; from 400 she is the king.
- Gone: `queen_alone_` and `QUEEN_SEED_UNITS`.

Simulator (pearls 30× sparser; indicative only):

- 32×16, 3 founders, against the previous build: our swarm went $7 \to 25 \to 29$ dragons (rounds 75, 100, 125), while the other side, with a non-splitting queen, fell to 1. The game still went to the other side on queen length: ours died and theirs lived, which shows how much the first tiebreak weighs.
- 60×46, 4 founders: she split until round 75, then walked her cycle on 86 of the next 100 turns, and was alive when the other team was wiped out.

45 When the queen is certainly dead, she saves the rest of her length

A split leaves the queen where she is, facing the same way: it cannot save her, only her length. So she splits only when nothing she can do saves her (`play_split_escape`). Every candidate move must be one of:

- a head trade;
- a move that leaves her head no free side (unless it eats a pearl, so the split can come next turn with one more segment);
- a move from which the survival search, run to completion, finds at most one more move.

Unlike other dragons, an enemy that could reach her does not count as certain death, since it may not come. When she is doomed:

- the rear $L - 2$ segments become a new dragon, if its end is not boxed in too;
- she, the 2 left behind, dies as she would have anyway;
- the "cheap way out" room estimates cannot veto it, because her test already searched every way on.

Pocket raids stay off for her, and trap pricing still treats a trap as her death.

Test (a test build that starts her cycle at round 1, in the simulator's rich economy, where she gets boxed in):

- at length 102 she split off 100 (the child had 120 tiles of room);
- at length 54 she split off 52 (72 tiles);
- each time her head had no room left, and she died the next turn.

Found while testing: seeding is risky for her. As a founder she splits at length 4 and keeps 2, and she joins fights like any worker. In three rich-economy games she died at length 2, boxed in among enemies, by rounds 26–39, in this build and the previous one alike.

Part VIII
v22 (continued): the centre rush on small maps

46 Which maps

Top teams head for the centre on Autarky (54×18), Devil, Portals, Prisoners Dilemma and Tower Defense (32×16), Stripes (24×12) and Weakhold (40×15), but not on the larger maps. Area does not separate them: Trophy (625) and Queen of Spades (875) are smaller than Autarky (972) and are not on the list. **The short side does**:

- the listed maps' short sides are 12–18;
- every other mid-size or large map's is 24 or more;
- the tiny maps (Arena 11×11, Small 16×8, Colosseum and Default Small 16×16) fall under the same rule.

So the rush is on when $\min(W, H) \leq$ `CENTRE_MAP_MAX_SHORT_SIDE` $= 20$. The bot knows W and H from its first turn.

Added: compact maps. The centre also pays on Trophy (25×25, whose centre beds are 3.8× richer than its edges). Its short side is 25, the same as Queen of Spades (25×35), which is not on the list. Area cannot separate them either: Autarky (972) is on the list and larger than Queen of Spades (875). So a second condition was added: the rush is also on when $\max(W, H) \leq$ `CENTRE_MAP_MAX_LONG_SIDE` $= 30$.

- **On**: Arena, Small, Colosseum, Default Small, Stripes, Devil, Portals, Prisoners Dilemma, Tower Defense, Weakhold, Trophy, Autarky.
- **Off**: Queen of Spades (both), Default (32×32), Maze, Stronghold, Trauma, Slithery Fight, Islands, Schooltime, and the 64×64 maps.

It is not about pearls. On Prisoners Dilemma, Weakhold and Autarky the centre's pearl beds give 2–9% of the outer rate; Tower Defense's give 15× it. On every listed map the teams start at opposite ends of the long axis, so the centre is the ground between them.

47 What it does

- For the first `CENTRE_RUSH_END_ROUND` $= 75$ rounds, every dragon gets `CENTRE_PULL` $= 300$ (thousandths of a segment) per step a move brings it closer to the centre. 450 was tried; 300 was kept.
- Exploring is worth 180 a step plus up to 120 for dispersal, so the centre wins over exploring. A pearl next to the dragon still wins over the centre.
- "Closer" is by path: a search from the centre over the neighbour table each turn of the rush (`prepare_centre`), with kelp and portals as we know them and unseen edges open. A straight-line pull would press dragons against walls.
- The pull stops within `CENTRE_RADIUS` steps of the centre (short side / 4, at least 2), so dragons do not pile onto one tile. It applies to every goal but fights and feeding (`goal_progress` now adds it to `goal_progress_base`).
- After round 75 it is off, so nobody gets stuck in the middle.
- The queen walks her cycle and does not rush (her moves are her own; see the queen part).

48 Simulator check (random maps, pearls stretched 30×: indicative only)

`probe_centre.py` gives the mean head distance to the centre (rush build vs current build):

- 32×16, 3 founders each: 4.0–5.7 vs 7.5–14.0 from round 25 to 75. Afterwards the rush build spreads out again (11.0 at round 100).
- 40×15, 4 founders each: 4.6–7.2 vs 11.8–16.6 from round 10 to 75.
- Both games, and the second with the sides swapped, left the rush team with far fewer dragons by round 75 (4–5 against 8–15). The likely reason is fewer pearls eaten on the way, so fewer splits. Whether this holds on the real maps, where top teams do rush, is for the sandbox to show; `CENTRE_PULL` is the knob to try lower (around 150, so pearls compete).

Part IX
v22 → v23: per-map settings and the queen's survival

v23 starts here. Everything before this part is v22.

49 A map table replaces the size rule

Autarky (54×18) turned out to be a near-certain loss with the centre rush, while Trophy (25×25) gains from it. No size rule fits that, and neither does a pearl one: Autarky's centre beds give 3% of its outer rate, but Weakhold's (2%) and Prisoners Dilemma's (9%) are just as poor, and the rush pays on both. So the choice is now made per map:

- `Config::CENTRE_MAPS` lists the sizes with the rush: 11×11 (Arena), 16×8 (Small), 16×16 (Colosseum, Default Small), 24×12 (Stripes), 32×16 (Devil, Portals, Prisoners Dilemma, Tower Defense), 40×15 (Weakhold), 25×25 (Trophy). The bot knows the size from its first turn.
- Among the 25 official maps, maps of the same size all get the same answer, so the size identifies the map well enough. Should two of the same size ever need different settings, a founder can also tell them apart by its starting tile.
- **Any other size gets no rush**: Autarky, the larger maps, and any map we have never seen (the grand finals may use new ones).
- `CENTRE_MAP_TABLE_ENABLED` = `false` turns the rush off on every map.
- Gone: `CENTRE_MAP_MAX_SHORT_SIDE` and `CENTRE_MAP_MAX_LONG_SIDE`.

Checked in the simulator:

- 54×18: the table build never came within 4 steps of the centre. The size-rule build reached a mean of 4.6 steps by round 50, with 4 dragons in the middle.
- 32×16: both sides rushed as before.

50 Walled in, or only contested?

What happened (from two sandbox screenshots). A 2-long worker heading down had three moves:

- **right**: into a tile with kelp above and to its right, whose only open side was the head of an ally worker heading left;

- **down**: into the tile that worker was about to enter;
- **left**: open water.

It went right. The ally moved on, so its neck sealed the last side, and our worker died on its next turn whatever happened.

Why. The survival search counts every tile another head could reach on its next move as taken for our next step. With two heads nearby, that took every tile beyond left and beyond down. So all three moves came out "searched completely, 0 of 4 moves". Left and down were judged exactly as dead as the kelp corner, and a small difference in goal progress picked the corner.

Change. A move whose search says dead is searched again (`survival`) with only real walls: kelp, bodies as they drain away, our own body, but not other heads' reach (`wall_free_at_`, built before those marks in `prepare_survival`).

- **Still dead**: `SurvivalResult::walled`, copied to `Candidate::walled`. It is priced as certain death (`trap_cost()` in full, not 0.9× the shortfall), so it can never be preferred to a move that is only contested.
- **Through**: the move is *contested*. It takes the second search's result (a way on), and the usual threat prices (adjacent heads, reach for valuable dragons, ally congestion) judge the contest.
- **Second search not run** (out of nodes or points): the first result stands, as before.

It runs only on moves the first search rejects, so it costs little. The queen's "certainly dead" test uses the same results, so a merely contested queen no longer counts as doomed.

Test (`sim/scen_walled.py`: the screenshot position rebuilt tile for tile; our worker's candidates):

Move	Before (search, score)	After (search, score)
right (kelp corner)	0 of 4, complete; −5514	0 of 4, walled; −6014
down	0 of 4, complete; −5175	4 of 4, safe; −675
left	0 of 4, complete; −4995	4 of 4, safe; −495

Before, the three were within about 500 points of each other. Here left happened to win; in the real game, right did. Now the dead end trails by about 5,500. A full simulator game against the previous build: no crashes, no timeouts.

51 Kill the enemy queen on sight

Their queen's length is the first tiebreak. So, in `queen.hpp`, every dragon checks this before anything else it might do (`play_queen_kill`, first in `play_turn`):

- **Who**: any of us that sees the enemy queen (an enemy head with raw id 0 or 1) in its 7×7 view, except our own queen, who would die with her.
- **How**: our head onto hers kills her (head on head: we die too). Dragons move one at a time, so where we see her head is where it is when we move. The shortest way there this turn goes through tiles we can see with no dragon on them, at most `max_turn_steps` (free steps, then paid ones, keeping 2 segments). It is replayed exactly (`simulate_path`). If one exists, that is the move, ignoring roles, trade values and rounds. That includes the old "no aggression before round 50" rule, and kings and growers.
- **Out of reach**: `QUEEN_HUNT_PULL` = 1000 per step closer to her head, whatever the goal (`goal_progress`).
- `KILL_ENEMY_QUEEN` = `false` turns both off.

Scenario test (`sim/scen_qkill.py`, round 10): their queen died head on head in all three cases.

- adjacent: one step;
- a 6-long worker two tiles away: one step after she moved towards it;

- a 3-long worker three tiles away: two steps, one of them paid.

Simulator games (seed 6, 4 founders a team, pearls 30× sparser; `probe_qkill.py`). In this simulator both teams' founders start next to each other, so a founder of ours traded itself for their queen on round 1. Both sides then had 3 dragons, but in both games (sides swapped) the killing team fell behind and was wiped out:

- game 1: 12 against 26 dragons by round 60;
- game 2: 11 against 21;
- control without the rule: 18 against 17.

Two games, a simulator with the wrong economy and starts unlike the real maps (where teams begin at opposite ends): to be checked in the sandbox.

52 The official maps in the simulator

`sim/engine.py -map NAME -seed S` now plays any of the 25 official maps as the official engine does (`config.cc`, `pearls.cc`):

- **Map contents**: kelp, portals, starting dragons in file order (ids 0 and 1 are the queens), rounds from 0, the map's unit limit (none of the 25 sets one: 64).
- **Pearl beds**: a bed and its mirror share one countdown. The gaps are drawn with the engine's own generator (`std::mt19937_64` seeded with the match seed), so with a replay's seed every pearl spawns on the same round as in that match. Only a spawn blocked by a dragon on the bed can differ.
- **Checks**: every map loads, kelp mirrors on all of them, every portal crossing leads back, and the generator matches the C++ standard's reference value.

Two lost games reproduced (v23 against itself, so where our pearls go rather than how they play; (dragons, total length)):

Map, seed		round 25	round 50	round 75	round 100
Schooltime, `0xa5e0c0ddde697915`	top team	(5, 11)	(7, 18)	(10, 26)	(22, 52)
	v23 before	(4, 8)	(13–15, 34–37)	(23–24, 62)	(20–34, 54–84)
	v23 now	(5, 11)	(16–18, 41–43)	(24–30, 62–74)	(33–35, 85–87)
Autarky, `0xfd11114afed14603`	top team	(12, 30)	(8, 19)	(16, 38)	(15, 38)
	v23 now	(12, 33–36)	(12–14, 29–34)	(9–12, 28–32)	(14–15, 38–41)

Our pearl economy keeps up with theirs on both. What lost Schooltime was the queen (next section).

53 Boxed in: keep the part that can live

What happened.

- **Schooltime**: each queen starts 4 long, sealed in a 2×2 kelp box, so every move hits her own body. Ours died on round 0. The top team splits off the 2-long tail; that child is boxed too and dies, and the queen eats the pearl it leaves. She then lives 3 long, circling the box all game. Their (5, 11) at round 25 is exactly that. Since the queen's length is the first tiebreak, the match was lost on round 0.
- **Slithery Fight**: the queen starts coiled with no free side. The split escape always kept 2 (the v13 rule: the rear $L-2$ leave, the head part dies), and on round 1 the remnant rule made her die on her own neck.

- The same rule cost a unit in most split escapes: the 2-long remnant died even where it could have followed the child out.

Change (`endgame.hpp`, `choose_escape`). The child moves later the same round:

- its first step frees its tail, our segment `keep`;
- a child that cannot move at all (boxed, and under 4 so it cannot split) dies at once and frees all of its tiles.

When such a tile is beside our head, we live on as well. The split keeps the most length that does this:

- Candidate keeps: each body segment beside the head, $L-2$, $L-3$, and 2 (the old escape).
- A plan wins on length saved: ours if we live, plus the child's if it has a way out.
- The queen (the part we keep) must live first.
- A child that dies at once is a price paid only when our head is otherwise certainly dead.
- All earlier gates (`fine`, room, tail threatened) still come first.

Supporting changes:

- **The child holds** (`main.cpp`, `escape_child_holding`). A child that got its parent's body handover makes no ordinary split for its first `ESCAPE_CHILD_HOLD_TURNS` $= 3$ turns (a split would leave its tail where it is). On its first turn, a move that leaves its tail in place (eating with every step) costs `ESCAPE_CHILD_STALL_COST` $= 5000$. The handover now carries the child's segments whatever we keep.
- **Remnant clear** (`play_clear_remnant`): only when boxed in (no free side), and never the queen.
- **Ordinary splits check the parent** (`split_parent_ok`). The child moves right after the split, from our tail, and may take a free side of our head next to it. We need a free side away from the child, or two beside it. On Slithery Fight the queen's founder split on round 1 had just one, and the child took it.
- **Our queen's last ways out** (`queen.hpp`, `queen_exit_block`). She moves first each round. A worker's move onto a free side of her head, while she has at most two, costs `QUEEN_EXIT_BLOCK_COST` $= 4000$.

Checked (round-0 to round-2 traces):

- **Schooltime**: split 2, the child dies, she eats its pearl and circles the box at length 3. Both queens are alive at rounds 60 and 200.
- **Slithery Fight**: she keeps 7 (the child of 18 leaves first), follows it out, and seeds as usual. Before, she died on round 1.
- **Our deaths in rounds 0–3** (team A), before and now:
  - Help 1 and 0;
  - Unsw 3 and 2;
  - Autarky 3 and 2;
  - Slithery Fight 8 and 4.

  The escapes that still lose their remnant have no body segment beside the head, e.g. a straight body nosed into a dead end.
- **Small**: still lost on round 3. She starts in a 1-wide corridor that runs into the corner where two of our starting dragons are, and that worker had only two moves, both next to her.

54 Farming good beds

A bed's respawn rate is learnt from the countdowns seen on it (`bed_gap`). The countdown runs $g, g-1, \ldots, 1$ after each spawn, so its mean over time is about $(g+1)/2$. With `GOOD_BED_MIN_SAMPLES` = 3 rounds of it seen, a bed whose estimated mean gap is at most `GOOD_BED_GAP` = 12 is *good*. A good bed is:

- an objective `GOOD_BED_SOON_ROUNDS` = 8 rounds before it spawns (other beds: 4);
- once its timer has run out while nobody watched, an objective at cost `GOOD_BED_EXPIRED_COST` = 2 for up to `GOOD_BED_EXPIRED_MAX_AGE` = 150 rounds (other beds: 5, for 40);
- worth 80 instead of 50 in `region_value`;
- free of `VISIT_COST` when walked over again (farming is not wandering in circles).

Effect (official maps, 100 rounds, new build against the build without it, sides swapped). Bed pearls eaten, both games:

- Devil 100 against 76; Schooltime 231 against 198; Portals 121 against 110;
- Autarky and Default even; Weakhold not changed at all.
- Overall +12%.

The limit is memory: every split starts a new process that knows nothing, and most dragons are young.

Where the rest goes. Measured yield is far below what the beds give:

- Devil's 18 one-round beds give 1,800 pearls per 100 rounds; we eat about 50.
- They lie in 1-wide vertical tubes, and Weakhold's and Stripes' good beds sit in dead ends.
- On Portals, most 2×2 bed blocks cannot be reached at all, and the rest are 1-tile dead ends behind portals.

Farming those needs a dragon that stays and loops through a tube, a bigger change than this one.

55 Pocket discipline: not changed

Raids already need 3 pearls inside the pocket (`POCKET_MIN_PEARLS`), and the keep-the-most escape above now saves the unit when a pocket is 2 wide. In a 1-wide dead end the head part still dies, as before. No new rule: the risk of dragons refusing pockets outweighs what is left.

56 v23 against the morning's v23 (simulator, official maps, 200 rounds, sides swapped)

Map	New as A	New as B
Schooltime	won (queen 3 against 0)	won (queen 3 against 0)
Slithery Fight	won	won
Weakhold	won (queen 14 against 0)	won (queen 6 against 0)
Help	lost	won
Devil	won (wiped them out)	lost (wiped out)
Autarky	won (queen alive)	lost

8 of 12 by the official ranking. Devil is a wipe-out race whichever side starts first. No crashes or timeouts.

57 The queen stays out of fights

Why. In self-play 27 of 50 queens were dead by round 100, almost all 2–3 long while seeding. In real games she also:

- crossed a portal whose far side held a dragon, and died at once;
- was boxed in by our own dragons.

Changes.

- **Valuable from round 0** (`compute_roles`). Before, she was valuable only from round 75, when she starts keeping her length. Now she has the king's caution throughout:
  - enemy reach threats rule moves out;
  - crowd and unseen-head risks are priced;
  - a portal whose exit we cannot see now is never a safe move;
  - no hunting and no head trades (`eligible_target`);
  - escape sprints are allowed.
- **Her death is priced high** (`death_cost`): at least `QUEEN_DEATH_LENGTH` = 30 segments, whatever her length. Every risk is priced against it. A blind portal now costs her about 9,600 points; at length 3 it was about 1,500.
- **No fight goal** (`prepare_fight`) and **no centre rush** (`centre_pull`): the middle is where the fights are.
- **Away from enemy heads**: `QUEEN_ENEMY_DISTANCE_STEP` = 150 per step from the nearest one (workers: 12).
- **Room from our own**: a worker's move onto any free side of her head costs `QUEEN_SIDE_COST` = 600, and 4,000 when she has at most two free sides (`queen_exit_block`).
- **She keeps 4 while seeding** (`QUEEN_SEED_KEEP`). At 2–3 long she could neither escape-split nor outlast a crowd. She now splits only from 6 long, giving the child everything above 4.

Queen deaths by round 100 (self-play, all 25 maps, seed 1, 50 queens):

Build	Queens dead
before (this morning's split fixes and farming only)	27
cautious, still splitting down to 2	18
cautious, and nobody hunts queens (control)	15
cautious, keeps 4 while seeding (this build)	13

Still lost:

- **Small**: both queens on round 3, as before (her corridor runs into a corner our starting dragons fill).
- **Prisoners Dilemma**: both, boxed at length 3. She starts 3 long, so she cannot split there.
- The rest are scattered head-on deaths.

The portal and own-team cases are covered by these rules; no scenario was rebuilt for them.

Against the previous build (8 official maps, 200 rounds, sides swapped): won 11 of 16. Her queen was alive at the end in 11 games; the old one's in 4.

- **Seeding is unaffected**: 301 against 305 dragons at round 50, summed over the 16 games.
- **Devil** went against us twice. Four more Devil games split 2–2, so that looks like a wipe-out race with no clear favourite rather than the change.

- Some wins were by the queen alone, with a smaller total (Default 8 against 67, Weakhold 24 against 76).

Part X
v23 → v24: portals

v24 starts here. Everything before this part is v23, which ended with the queen staying out of fights (section 57).

58 Portals: rooms, pockets, and pacing in front of them

Measured (`sim/portal_loiter.py`; v23 against itself, official maps, seed 3, 150 rounds). Two separate faults:

- **Shut in a portal room.** Default's centre (a grid of 4×4 rooms), Autarky's 3×3 boxes and Portals' cells are sealed by kelp; the only ways out are portals on their inner edges. Leaving means a crossing whose far tile we cannot see, priced at 30% death (`P_BLIND_PORTAL`). A worker that had eaten what was there circled the room instead: on Autarky four dragons for 108–142 rounds each, on Default one for 124.
- **Pacing in front of a portal.** The searches behind the goals (pearl route, objective map) treat a crossing as an ordinary step, but the move prices it as a 15–30% death. So the goal led a worker up to the portal, the crossing lost (e.g. −1460 against −562 for turning away), the goal led it back. Rounds a dragon spent pacing (at most 4 tiles in 12 rounds) near portals: 125 on Portals, 24 on Islands. Away from portals: 9.
- Most such pulls on Portals came from **pearls in portal pockets**: 1-tile cells whose only side out is a portal. A small dragon that goes in eats and dies there: its neck is in the portal and it is too short to split out.

How risky blind crossings really are (`sim/portal_exit_risk.py`, share of rounds a crossing's far tile held a dragon):

- rounds 0–49: 2–14%;
- rounds 50–149: 8–19%;
- later: 12–25%.

`P_BLIND_PORTAL` = 0.3 is pessimistic but not far off late in a game. It is left as it is for the fixes in this section; section 59 then prices crossings by round.

Changes (`main.cpp`, `scoring.hpp`):

- **Shut in a room** (`update_portal_room`, once a turn). We are shut in when:
  - the tiles we can reach without a portal (at most `PORTAL_ROOM_MAX_CELLS` = 64) are all seen;
  - they hold no objective at all;
  - a portal leads out.

  Then a crossing (not onto a body we know of) is priced at `P_ROOM_EXIT` = 0.1 and is worth `ROOM_EXIT_VALUE` = 1500. A valuable dragon may then cross blind too (its safety filter allows it), and a portal we never looked through counts as exploring. Not the queen: a room is a fine place for her.
- **Portal pockets are not objectives** (`portal_pocket_worth`). A pearl or bed whose portal-free tiles are few (at most `PORTAL_POCKET_MAX_CELLS` = 8), all known, and a dead end or too

small to turn in, is left off the objective map. The exception is when a pocket raid is allowed and it holds `POCKET_MIN_PEARLS`, as for kelp pockets.

- **A portal on the way is crossed or dropped** (`refuse_route_portals`). The crossing's candidate records whether it makes progress on the goal (`route_portal`). If we then choose another first step, its far tile is marked a trap for `PORTAL_REFUSE_ROUNDS` = 15, so the route and objective searches go round it and the goal picks something else.
- The objective map now records which tiles are objectives themselves (`objective_src_round_`).

Results (same probe):

	Before	After
Autarky: stays ≥20 rounds by a portal	7 (four of 108–142)	2 (26 and 22)
Default: stays ≥20 rounds	15	1
Portals: rounds pacing near portals	125	5
Islands: rounds pacing near portals	24	1

Against v23 (8 portal maps, 200 rounds, sides swapped): 9–7.

- Maze, Prisoners Dilemma and Stripes were decided by side, not build: mirrored results, or the A side won every game (Stripes, 4 more seeds: 4–4).
- Total length 1,198 against 1,362, most of it one Schooltime game (93 against 393).

Points (metered wasm, 12,345 recorded turns on Unsw, 64×64): median 17.3M → 17.5M, max 43.3M → 43.6M.

59 Open early, only good ones late

Many portal maps keep pearls on the far side of their portals, and early in a game whoever gets through first eats them. The crossing risk was 30% at any time; the simulator measured 2–14% in rounds 0–49 and 12–25% after round 150 (section 58). So:

- **A blind crossing's risk follows the round** (`blind_portal_p`): `P_BLIND_EARLY` = 0.08 up to round `PORTAL_RISK_EARLY_ROUND` = 40, rising linearly to `P_BLIND_LATE` = 0.20 at `PORTAL_RISK_LATE_ROUND` = 150. Half that when we saw the far tile free in the last 4 rounds (was a flat 0.1). The constants `P_BLIND_PORTAL` and `P_SEEN_FREE_PORTAL` are gone.
- **The early window is longer**: `PORTAL_EXPLORE_END_ROUND` 80 → 100. Until then, as before, workers treat a portal they have never seen through as an exploration objective and an ordinary move (5% risk).
- **Late, only good portals.** From round 100 the far side is valued by the pearls and timers we know of there (`region_value` with unexplored tiles at 0): `PORTAL_GOOD_PER_PEARL` = 300 per expected pearl more than around us, up to `PORTAL_GOOD_CAP` = 2500.
  - Nothing for a far tile with no way on (a pocket, a dead end), unless it is a planned pocket raid.
  - Before round 100 the far side keeps its old small weight (48 per pearl, unexplored tiles counted, ±360).
  - For a 3-long worker at round 150 the crossing's risk is about 1,000, so it needs about 3 more known pearls on the far side; a longer dragon needs more.

Tried and dropped (each against the build with only the portal fixes above, 200 rounds, sides swapped):

- A bonus of 720 for an early crossing the goal's way goes through lost all 4 Default games (total length 68 against 238).

- Valuing the far side at 500 per pearl from round 0, unexplored tiles counted, went 4–8 and lost all 4 Default games. On Default's grid of portal rooms both sent the swarm hopping room to room in the middle of the map.

Results.

- **Against the portal fixes alone** (6 maps, sides swapped): 7–5, total length 1,479 against 1,099. Help, Islands and Portals went to the B side in both orders.
- **Early crossings** (rounds 0–99, seed 5, the team that crossed more): Portals 12 → 53, Default 28 → 78, Islands 58 → 111.
- **The cost**: more small dragons die finding out where an unknown portal leads. On Portals many lead into 1-tile pockets, where a 2–3-long dragon eats and dies. Deaths on a crossing move, early: Portals 4 → 11, Islands 7 → 23.

v24 against v23 (all of this part against the build that ended v23; 6 portal maps, 200 rounds, sides swapped): 5–7, total length 1,275 against 1,201. Seven of the 12 were settled by whose queen was alive or longer, not by length: Autarky as A had 122 against 17 and lost on the queen (4 against 11). Read it as even. Since the queen tiebreak swamps everything else in these runs, the portal changes are best judged by the loitering and crossing figures above, and by sandbox games on portal maps.

Part XI
v24 → v25: dispersion

v25 starts here. Everything before this part is v24.

60 Spread out early

The problem. We keep losing early eliminations, and losing consistently means exploration is not luck. Several of our workers going to the same spot gain little, since one or two there split quickly and fill it. They should spread over the map instead.

What was there already:

- **Pearls**: each pearl is left to whichever known teammate reaches it first (`pearl_taken_by_ally`: visible teammates by path steps, others by the position they last sent, at most a round old).
- **Destinations** (v14, `explore.hpp`): every worker heads for a part of the map nobody has explored, chosen by its id. Only on maps of 1,600 tiles or more and from round 20, so not on most official maps.
- **Exploring**: nothing shared it out. Every never-seen tile within 30 steps was an objective for everyone, so dragons near each other headed for the same unexplored ground, and on the centre-rush maps every worker ran to the middle.

Measured (`sim/spread.py`, v24 against itself): teammate heads within 6 tiles of each head, against what an even spread over the map would give.

- Trophy, round 50: 9.6 against 4.3.
- Schooltime, round 75: 3.7 against 1.6.
- Default, round 50: 3.3 against 1.8.

Changes (`main.cpp`, `scoring.hpp`):

- **Shared exploring** (`ally_claims`, in `build_objective_map`). Until `CLAIM_END_ROUND` = 200, a never-seen tile, or a timer that ran out while nobody watched, that a known teammate reaches first costs `CLAIMED_EXTRA` = 10 more (a never-seen tile costs 8).

  - Known teammates: those we see, by their path steps (`ally_etas_`), and others by the position they sent in the last `CLAIM_SIGNAL_AGE` = 4 rounds plus that age.
  - Ties go to the lower id.
  - It is a cost, not a ban: with nothing else to do, a dragon still explores ground a teammate is nearer to.

- **At most two rush the centre** (`centre_crowded`). On the centre-rush maps a dragon with `CENTRE_MAX_RUSHERS` = 2 or more known teammates closer to the centre gets no centre pull.

Tried and dropped (each against v24, the same 6 maps, 150 rounds, sides swapped):

- **Destinations on every map from round 0** (the v14 rule without its size and round limits): 3–9, and 24% less length at round 50 (22.8 against 30.0). Sending workers away from food they can see costs more than it explores, as v14 found.
- **Stronger shared exploring** (extra cost 20, positions up to 6 rounds old): 6–6, against 8–4 for the settings kept.

Results against v24 (12 official maps, 24 games, 150 rounds, sides swapped; `sim/ab.sh`, `sim/ab_econ.py`):

- **Games**: 14–10 by the official ranking.
- **Wipe-outs**: we wiped them out 6 times; they wiped us out once.
- **At round 50**, on average: 9.7 dragons and 24.5 length, against 8.4 and 22.2.
- Several of the 10 losses were on the queen tiebreak with more length (e.g. Portals as A, 47 length against 39 at round 100).

Points (metered wasm, 11,952 recorded turns on Unsw, 64×64): median 17.4M → 18.5M, max 46.7M → 46.8M.

61 Anticipated pearls

Every tile in view shows how many rounds until its next pearl, and the bot already remembered that countdown (`CellMemory`). Until now it only used it within 4 rounds of the spawn (10 for the king), at cost rounds left +1 plus the walk. So a dragon passing a bed knew exactly when it would fill, and forgot about it until it was almost due.

Timed beds on the official maps (beds that cannot spawn before round 20 or later):

- **Prisoners Dilemma**: the two middle columns (32 beds) all spawn once, around rounds 479–489. 44 beds either side of the middle first spawn around 150–250, then every 150–250 rounds.
- **Tower Defense**: 38 beds (centre and corners) spawn once, between rounds about 399 and 499.
- **Unsw**, **Maze**: 36 and 28 beds at 250–500. **Australia**: 126 beds starting after round 40 or 100. **Schooltime**: nothing before round 19.
- **Queen of Spades But She Ages**: its 170 delayed beds (rows timed 40–80 at the edges up to 400–480 near the middle) are all sealed off by kelp (`sim/engine.py` reachability from every start). They spawn but nobody can eat them, exactly as on the normal Queen of Spades, where the same tiles are 1–1000 beds.

Change (`main.cpp`, `spawn_forecast_cost`). Any map, no map table:

- **When to set off.** A known future spawn becomes an objective once its rounds left are at most our turns to get there plus `FORECAST_MAX_WAIT` = 2, so we arrive as it spawns.

- **Its cost** is that wait, max(0, rounds left $+1-$ our turns there); the objective search adds the walk. A pearl that is about to appear right where we can be in time ranks like one already there.
- **Bursts rank higher.** Each other spawn within `FORECAST_CLUSTER_RADIUS` $= 3$ tiles and `FORECAST_CLUSTER_WINDOW` $= 6$ rounds takes $1/$`FORECAST_CLUSTER_DIV` $= 1/3$ round off the cost, at most `FORECAST_CLUSTER_CAP` $= 8$.
- The objective radius bounds it: 30 steps for workers, 200 for the king.
- `SPAWN_SOON_ROUNDS`, `KEEPER_SPAWN_SOON_ROUNDS` and `GOOD_BED_SOON_ROUNDS` are gone.

Tried and dropped: making every known future spawn an objective at once (cost = the wait). Dragons went early and circled on the beds; a dragon's body on a bed when it spawns blocks the pearl. In a test copy of Prisoners Dilemma with its burst moved to round 80 (`sim/timed_beds.py`, 3 seeds, sides swapped) that build ate 59 of the burst's pearls against 79 for v25 without forecasting. Setting off on time turned it round: 77 against 60.

Results (against v25 without forecasting):

- **12 official maps**, 150 rounds, sides swapped: 14–10. At round 50, 9.0 dragons and 23.8 length against 8.5 and 23.0.
- **Prisoners Dilemma**, 4 seeds, full games, sides swapped: won 7 of 8. The delayed beds' pearls eaten: 126 against 59. In self-play these games end by wipe-out between rounds 196 and 277, so the round-480 burst itself never came.
- **Tower Defense**, 2 seeds: 2–2, all decided by the A side wiping the other out before the round-400 beds.
- **Points** (metered wasm, 23,150 recorded turns on Big Empty, 64×64, every tile a bed): median 18.3M → 18.4M, max unchanged at 29.1M.

Part XII
v25 → v26.0.0: a portal is a way out

v26.0.0 starts here. Everything before this part is v25 (a top-30 bot at that point). v26.0.0 is v25 plus the next section.

62 A portal we have never looked through is a way out

What happened (sandbox, Maze, round 61). The queen, 4 long, had followed a chain of portals early on and stood with her head in a small kelp pocket, facing the wall:

- **west**: a 1-tile pocket whose only other side is a portal she had never looked through;
- **east**: a 2-tile dead end;
- **behind**: her body, back the way she came.

She split (her rear 2 escaped the way she came), went east the next round, and died in the dead end.

Why. The survival search (`surv_dfs`) treated a portal whose far end we do not know as a wall: it cannot see where it leads.

- West scored 0 more moves and east 1. Both are "hopeless" by the queen's certain-death test (every move a head trade, no free side, or a complete search allowing at most one more move), so she made her escape split.
- With only her neck beside her head, nothing could be kept alive (section 53): she kept 2 and her rear 2 left.

- At length 2 the same search preferred east (1 more move) to the portal (0), and she died there.

The crossing itself was most likely safe: a portal's far tile held a dragon in 2–25% of rounds in the simulator (section 58).

Changes (`survival.hpp`, `scoring.hpp`, `endgame.hpp`; `SurvivalResult` and `Candidate` carry it from `main.cpp`, `route.hpp`, `explore.hpp`):

- **A way on.** In the survival search, a portal beside the head whose far end we have never seen counts as reaching the target, but only when no other move from there does (it cannot see beyond it). The result says so (`via_portal`).
- **Priced as a crossing.** A move whose only way on is such a portal pays the crossing's chance of landing on a body: 5% for workers in the early portal window, otherwise `blind_portal_p()` (8% early, rising to 20% by round 150), times our death cost. For the queen, whose death counts as at least 30 segments, that is about 2,600–6,600 points: far less than a dead end, far more than open water.
- **Not hopeless.** A step through such a portal is not "certain death" for the queen's test nor the workers': its exits are unknown, not zero.

The room estimate (`timed_space`) already counted such a portal as open, so the two searches now agree.

Checked.

- **The screenshot rebuilt** (`sim/scen_maze_queen.py`: the pocket, a portal to the far side of the map, round 61). v25: split, east, dead on round 64, exactly as in the sandbox. v26.0.0: west, through the portal on the next move, and still alive and 4 long at round 66.
- **Queen deaths by round 100** (self-play, all 25 maps, seed 1, 50 queens): 18 → 16. Most of the rest are head-on deaths to the other side's queen hunters, which self-play doubles.
- **Portals** (`sim/portal_loiter.py`, seed 5, 200 rounds, both teams): deaths on crossing moves Portals 49 → 43, Default 8 → 11, Maze 1 → 9. On Maze, crossings went 54 → 156 and both teams ended longer (123 and 101 → 127 and 136).
- **Against v25** (12 official maps, 150 rounds, sides swapped): 11–13, the same length at round 50 (33.0 against 33.3 averaged over both batches). A narrow fix, so no change expected there.
- **Points** (metered wasm, 11,952 recorded turns on Unsw): unchanged (median 18.5M, max 46.8M).

63 Tried after this, and reverted

Everything below was built on top of the change above, and all of it is reverted. v26.0.0 is the change above alone; v26.1.0 (next part) was built afresh on top of it.

Why. The full build against v25 on the real engine (`unswbc run`, every map, sides swapped, 150 games): V25 won 78, it 71, with 1 draw. That gap is within the luck of 150 games, but nothing showed it was better. The changes had been tested mostly in the simulator, a few at a time, each against the build before it, on 8–16 games, and against our own bot, so each small step looked better, or no worse, without the whole being any better.

What was reverted (the full build and its changelog are kept, should any of it be worth another look):

- **The queen's deaths, traced**: no seeding or escape split inside an enemy's reach; she moves rather than split when no split keeps her alive; pearls on a hunter's way and enemy tails counted in her threat map; her route only to tiles with everything near them in view; free food on her route.
- **The whole-window rule** (after round 75, an enemy whose tail she cannot see covers its whole 7×7).

- **More early portal pull for workers** (crossing value, objective cost, cheaper blind exits, far-side pearls).
- **Two-sided portal pockets count as tunnels** (`portal_pocket_worth`).
- **Portals: tunnels and death cells**: one-way tunnels, walls and pockets in portal messages, a survival search that looks past a door.
- **No centre rush on Portals** (told apart by its one-tile portal cells).
- **No early portal exploring for the queen**.

Part XIII
v26.0.0 → v26.1.0: early portals

v26.1.0 starts here. Everything before this part is v26.0.0 (v25 and the door fix, section 62).

64 Early portals: workers go in, the queen stays out

The aim. Early on, a worker that sees a portal goes through it; the queen does not. The king and the growers do not either. Nobody paces in front of a portal.

What happened (sandbox, Portals, round 7). A worker at (28,7) had three portals in plain sight and took none of them. Other workers did the same on other maps.

Why. Five things held workers back:

- **Only the explore goal paid for a crossing.** The crossing value (180) was paid only under the explore goal. A worker with any other goal (eat, hunt, fight) got nothing for going through.
- **Eat called it a detour.** Under the eat goal, a portal we have never looked through counted as a step off the route to the pearl.
- **The objective map charged for it.** Each unknown portal on the way cost 5 there.
- **The centre rush won.** On 32×16 and the other centre maps, it pulls 300 a step towards the middle, more than the crossing was worth.
- **Fights through walls.** Fights were picked by straight-line distance (at most 4), walls ignored. On Portals, workers lined up along the double wall between the two halves, "fighting" enemies they could never reach, right beside the portals they should have taken. In the simulator probe, every skipped portal beside a worker was under the fight goal.

Changes (`main.cpp`, `scoring.hpp`, `queen.hpp`). Workers, while `portal_exploring()`: not retaining length, not valuable, before round 100.

- **Crossing value.** `PORTAL_EXPLORE_VALUE` 180 → 800, whatever the goal (`portal_value`).
- **Not a detour.** Under the eat goal, such a portal is no longer a step off the route.
- **Cheaper on the objective map.** `UNKNOWN_PORTAL_EXPLORE_COST` 5 → 2 (an unseen tile costs 8).
- **The centre rush gives way.** It pulls nothing while a portal nobody has been through is in view (`note_unexplored_portals`, `centre_pull`). It is otherwise unchanged; removing it for the centre sizes was not needed.
- **Fights by path, for every dragon.** An enemy is near when it is at most 4 steps away by path. The path search goes round walls, through known portals, and ignores bodies, which move (`fight_path_distances`, `prepare_fight`).

- **Keep-out for the queen, the king and the growers** (`PORTAL_KEEPOUT`, `apply_portal_keepout`, `queen_safe`).
  - A move that takes a portal into the unknown costs a death while some other move keeps them alive. That covers a pair nobody has seen both ends of, a far tile out of sight, and a step whose only way on is such a portal.
  - "Keeps them alive" means: not a head trade, not walled, no hard threat, and a way on.
  - When every other move dies, the portal is still taken, as the way out (section 62).
  - A portal onto a tile in view is a step like any other. On Stripes, the box of beds next to the start is one. Strict keep-out lost Stripes 0–4 in all three runs that had it; with this exception it went 2–2.

Tried, not kept: sending a pair at once. A pair a dragon has just seen both ends of went out by sonar on the next round, rather than waiting its turn. The aim was to tell teammates sooner that the portal is explored, so they pick another. It made no difference: pairs crossed by round 50 were 3.6 either way, and by round 100, 5.2 either way. Left out.

Checked.

- **Exploration** (simulator, `sim/explore_cov.py`): the 14 maps with 4 or more portal pairs, seeds 1–5, both sides, against v26.0.0, 100 rounds.
  - Pairs first crossed: by round 25, 1.5 → 1.9; by round 50, 2.7 → 3.6; by round 100, 4.5 → 5.2.
  - By round 50: Portals 3.2 → 4.8, Default 4.4 → 6.2, Schooltime 2.5 → 4.2, Stripes 1.4 → 2.5, Dilemma 0 → 2.
  - Maze by round 100: 8.3 → 7.3, slightly fewer. Stronghold, Trauma and Australia: unchanged. Their portals are seldom in view early, which is a dispersion problem rather than this one.
- **Seen, then entered** (`sim/portal_behave.py`, `walkaway.py`, 150 rounds).
  - On Portals, workers next to a portal never looked through took it 31 times out of 32.
  - Pacing 8 or more turns in front of a portal: 0–2 a game on Portals, Default, Islands, Stripes and Maze.
  - The queen crossed only when she had a single move, or into the in-view box on Stripes.
- **Real engine** (`ubcnative/ubcrun`, built from the official engine sources; all 25 maps, both sides).
  - The same seed does not replay the same game: deadlines are timed. So 100 games are good to about ±10.
  - Against v26.0.0, 100 games each:
    - worker changes only: 57–43;
    - worker changes + strict keep-out: 54–46;
    - keep-out only: 53–47;
    - + fight by path, strict keep-out: 49.5–50.5;
    - v26.1.0 (keep-out with the in-view exception): 52.5–47.5.
  - **v26.1.0 against v25** (150 games, seeds 3–5): 76.5–73.5. Queens alive 43 against 41; total length 13,800 against 11,899 (+16%). Portals went 1–5 in those 6 games; 16 more Portals games went 8.5–7.5.

Part XIV
v26.1.0 → v26.2.0

v26.2.0 starts here. Everything before this part is v26.1.0 (early portals, section 64). Next: dispersion.

65 A worker's split takes 2 off its tail

The rule. When a worker splits, the child is the 2 segments at its tail (the engine's child always comes off the tail), and the worker keeps the rest: $L = (L-2)+2 = (L-4)+2+2$, and so on, over the next turns. Before, the worker kept 2 and gave the rest (at most 9). So a long worker (Slithery Fight starts workers of 5–7) left a 2 at the front and a long dragon facing backwards. One switch, `SPLIT_TWO_FROM_TAIL` (`main.cpp`, `worker_child_size`); `false` gives the old sizes.

Not changed: split escapes (the rear part reverses out of a dead end or a box, `endgame.hpp`), pocket raids that end in one, and the queen's seeding.

Tried first: every split, escapes included. 45–55 against v26.1.0 over 100 games. Queens alive 24 against 31; total length −14%. With escapes saving only the 2 at the tail, the rest of a doomed dragon died with its head, and pocket raids had to go. So the rule is for worker splits only.

Checked.

- Simulator, Slithery Fight, 60 rounds: every worker split (35) had a child of 2. Escape splits kept their sizes (2–5, and the queen's 18 at the start); the queen seeded 3.
- **Real engine against v26.1.0** (all 25 maps, seeds 1–2, both sides, 100 games): 53–47. Queens alive 29 against 25; total length 7,733 against 9,002 (−14%). Slithery Fight went 1–3 in its 4 games. Within the luck of 100 games on all three counts.

Part XV
v26.2.0 → v26.3.0: Leo's v27.2.0 merged in

v26.3.0 starts here. Everything before this part is v26.2.0 (section 65).

66 What Leo's bot changes

Leo's v27.2.0 is V25 plus about 750 changed lines (`main.cpp`, `scoring.hpp`, `endgame.hpp`, `queen.hpp`; the rest only reformatted). Most of it is experiments behind switches that are off. These five are on in his build:

- **Late army** (his idea 2). Half the workers, by a hash of the id, do not feed the king from round 380 to 440: they harvest, hunt and explore, skip the late rendezvous, and may keep splitting. His numbers: 0.511 against V25 over 3,200 games.
- **The queen and portals** (his safety A). From round 100, while the queen can reach 64 tiles or more without a portal, any portal move is unsafe for her and costs 40,000. His numbers: 202–198, "too small to measure".
- **The queen waits** (his v25.1 and safety B). Certainly dead, the queen splits off only 2 and keeps her place and the rest of her length, where V25 let her rear $L-2$ escape and she died. She also waits when every move is hopeless or ends in a certain enemy reach (or crosses a portal, as above) while her own tile is out of reach.
- **Ally box** (his v27.1 idea 1). A move that takes every free side left to a teammate's head costs 0.7 of that teammate's death; taking all but one costs 300. His numbers: 0.542 head to head

against his v27.0.0 (800 games).

- **Early trade** (his v27.2). Before round 50, a worker (not the queen, not valuable) may trade head-on into an enemy as long as it or longer while we have 3 units or more. His numbers: 0.580 head to head against v27.1.0 (800 games).

Off in his build, so not brought over: workers growing before they split, sharing where the enemy queen is, a cautious queen pricing, corridor and dead-end costs, worker reach, child room, the army fed near the queen, bolder portals.

67 The merge

The five are ported as they are, each behind its own switch (`main.cpp`, after the rest of `Config`): `KEEP_ARMY`, `QUEEN_NO_PORTAL`, `QUEEN_WAIT_SPLIT` with `QUEEN_STALL`, `ALLY_BOX`, `EARLY_TRADE`.

- `main.cpp`: `army_worker`, its split stop; `apply_queen_no_portal`, `queen_room_without_portals`, `path_crosses_portal`; the early trade in `eligible_target`.
- `endgame.hpp`: feeders and the rendezvous leave the army out; the queen's wait split and stall in `play_split_escape`.
- `scoring.hpp`: `ally_box_cost`, in `goal_progress`.
- `queen.hpp`: the header comment only.

How it sits with ours:

- **Portals.** Our keep-out (section 64: queen, king and growers stay out of portals into the unknown, all game) and his rule (the queen, any portal, from round 100 when she has room) both apply.
- **The door** (section 62). A step through a portal we never looked through is still not hopeless for the queen: she takes it rather than wait, except where his stall rule calls a portal move no better than waiting.
- **Splits.** Our 2-from-the-tail rule is for ordinary worker splits, his wait split is the queen's: no overlap. His army may keep splitting after round 380, in 2s like any worker.

Two of our own v26 changes got switches too, for the tests below: `EARLY_PORTAL_PULL` (the worker pull of section 64; `false`: V25's 180, explore goal only, objective cost 5) and `FIGHT_BY_PATH`.

The port checked. With our own v26 changes switched off and no deadlines (`BOT_UNLIMITED`, so a game replays exactly), the merge and Leo's build played the same moves on every turn of three simulator games on maps without portals, where the door fix cannot act: Default Small (1,485 turns), Weakhold (10,404) and Tower Defense (6,182).

68 Results

Real engine (`ubcrun`), all 25 maps, both sides. 100 games are good to about ±10.

- **v26.2.0 against Leo's v27.2.0** (seeds 1–2): 38–62. Queens alive 22 against 33; total length 6,625 against 10,480.
- **v26.3.0 against Leo's v27.2.0** (seeds 1–6, 300 games): 156.5–143.5 (52%, ±6). Queens alive 103 against 73; total length 25,293 against 26,295 (−4%). By seed pair: 47–53, 57.5–42.5, 52–48. Islands went 2–10 over the 12 games, the one map with a run like that.

Which change matters (head to head, v26.3.0 against itself with one switch off, seeds 1–2, 100 games):

switched off	score (with–without)	queens alive	total length
early trade (Leo)	60–40	31 / 26	8,728 / 7,691
ally box (Leo), seeds 1–4, 200 games	92.5–107.5	65 / 65	16,875 / 15,805
late army (Leo)	52–48	31 / 30	9,641 / 6,909
queen waits (Leo)	52–48	29 / 28	9,433 / 7,704
2 off the tail (ours)	52–48	32 / 26	8,251 / 8,500
fight by path (ours), seeds 1–4, 200 games	95–105	59 / 60	16,506 / 16,847
early portal pull (ours), seeds 1–4, 200 games	92–108	63 / 63	16,728 / 17,544

- **Early trade is the one that shows**: 60–40, ten points, in line with Leo's 0.58. Before round 50 our workers now answer the enemy workers that ram them, and more games are won by elimination (31 against 19 of the 100).
- **The ally box does not show here**: 92.5–107.5 over 200 games (46%, ±7). Leo measured +4 points in his bot; in ours it is no gain, perhaps a small loss. It stays on, as in his build, pending more games.
- **Our own early-portal changes do not show either**: the worker pull 92–108 and fights by path 95–105, over 200 games each. Like the ally box, inside the noise and leaning the wrong way.
- **The rest are inside the noise** (52–48), the army and the queen's wait with more length.
- **All three of those off together** (ally box, worker pull, fights by path; seeds 1–4, 200 games against v26.3.0): 99.5–100.5. So their leanings were noise, as a whole they change nothing, and v26.3.0 keeps all three. By map it is another story, 8 games each: Help 0–8 without them, but Dilemma 8–0, Islands 7–1 and Big Empty 7–1 without them. Worth a look when we come to exploration.

Part XVI
v26.3.0 → v27.0.0: eliminations

v27.0.0 starts here. Everything before this part is v26.3.0 (section 68).

69 Why v-top_team eliminates us

Where. Real engine, v-top_team against v26.3.0 (seeds 1–2, 100 games): 59–41. All 34 of its wins by elimination came on the 11 maps under 1,100 tiles (Colosseum, Arena, Autarky, Default, Default Small, Devil, Prisoners Dilemma, Small, Stripes, Tower Defense, Trophy). On the larger maps the games ran to round 500, but both queens were usually dead and v-top_team ended with 3–10 times our units (Help 53 against 2, Islands 28 against 3, UNSW 50 against 12), so it won on the longest dragon: the "mostly eliminated" loss.

How: a unit snowball. On Devil, Stripes, Autarky and Arena both teams eat about the same for 30–40 rounds, but v-top_team has twice our units by round 40 (Devil: 5 against 6 at round 10, 5 against 10 at round 40, 2 against 20 at round 60). It splits every dragon the moment it reaches 4, its queen too on small maps. Ours sat at 4–5 in the 1-wide lanes of Devil and Stripes (no split in a corridor at length ≤ 5, v17) and our queen held 4–6 segments. In the simulator the unit count at round 25 nearly decides the game (296 games): ahead, we win 89%; level, 58%; one behind, 42%; two or more behind, 9%.

Not the economy. Each bot playing alone (the other team removed at round 0, simulator, 100 rounds) eats about the same on most of these maps (Default Small 85 against 83, Devil 213 against 197, Dilemma 18 against 17). The exceptions are maps whose food sits in portal pockets and dead-end lanes: Stripes 39 against 147, Portals 187 against 270, Schooltime 164 against 298.

Their bot, one mechanism off at a time. v-top_team reads its parameters from a JSON argument, so each of its mechanisms can be switched off without touching its code (wrappers in its own folder). Our win rate against it, simulator, 13 small maps, seeds 1–4, both sides (104 games each):

v-top_team with this off	our win rate
nothing (control)	43%
its queen as a worker on small maps (protected there too, `Q_AREA 1`)	**70%**
head-on rams (`RAM 0`)	**66%**
splitting at 4 (`SPLIT_MIN 6`)	58%
the lane economy (`LANE 0`; with the newborn-split hold too, 57%)	54%
no split on a newborn's first turn (`NB_NOSPLIT 0`)	42%
ram one shorter, hunt pull, centre-biased exploring, centre rush, hunting our queen, traps, rival discount on food, threat avoidance (each alone)	38–43%

Why the rams work: the drops. In the first 60 rounds of 16 games it rammed 71 times, we rammed 30. Two in three of its rams had more of its heads than ours within 3 tiles of the collision (ours: one in eight), and it ate 87% of the drops, 2.1 pearls a ram, against 0.3 for us. It also ate most of the drops of *our* rams. By who had the nearer head (any dragon but the two):

nearest head	rams	drops eaten by us	by them
theirs	92	0.25	2.21
equal	19	0.37	1.74
ours	36	1.06	0.94

Even with our head nearer, half the drops went to them. Our pearl routes divided a pearl's worth by $1 + 0.08$ for every other dragon's segment within 2 tiles of it (`ROUTE_WORKER_CROWD`): a pile of drops is exactly that.

70 What v27.0.0 changes

Six switches, all in `main.cpp`'s `Config`:

- **The queen plays as a worker on maps under 1,100 tiles** (`QUEEN_SMALL_AREA 1100`; `queen.hpp` `worker_queen`). She splits at 4, trades, rushes and explores like any worker, and is not crowned. Small fix on the way: as our last unit she never trades herself for their queen (on Arena both teams start with one dragon, and both would be out: a draw).
- **Nobody hunts the enemy queen on those maps** (`KILL_QUEEN_MIN_AREA 1100`). On small maps their queen is often one of their swarm, and the hunt (1,000 a step towards her, and a kill whenever reachable) walked our workers into it. On top of "s5n" below: +3 points, 216 eliminations suffered against 240 (416 games).
- **Short dragons split in corridors too** (`CORRIDOR_SPLIT_MAX_LENGTH 0`, was 5).
- **A pocket raid may start through a portal** (`PORTAL_POCKET_RAID`). Stripes keeps its food in portal pockets of period-1 beds; alone on Stripes we ate 65 by round 100 instead of 37.
- **No crowd discount on a worker's pearl route** (`ROUTE_WORKER_CROWD 0`, was 0.08; the king and growers keep 0.25).
- **No blind portal crossing onto a teammate** (`ALLY_PORTAL_GUARD`): an exit we cannot see with a teammate reported (status packet, at most 2 rounds old) on or next to it is an occupied exit. Two of ours crossing one pair from both sides met head-on 1.6 times a game on the small maps (Default, both Queens of Spades, Autarky); now 28% fewer.

Files: `main.cpp`, `queen.hpp`, `scoring.hpp`, and `explore.hpp` (it carries the exploration tests below, all off).

71 Tried, and left off

Each is a switch in `Config` (off), tested on its own against v-top_team in the simulator: 13 small maps, seeds 1–16, both sides, 416 games each (±2.4 points). The exploration tests from the start of V27 (bed frontier, scouts, rich spots over sonar, magic 0x8) stay in the code, off.

variant	win	eliminated	eliminating
v26.3.0	32.7%	255	109
no corridor split block, alone	34.4%	259	113
queen as worker on maps < 1,100 tiles, alone	37.3%	233	122
queen as worker only when our start is ≤ 5 segments (`QUEEN_WORKER_MAX_START`), with no corridor split block ("s5n")	39.2%	240	131
+ their ram rule, sprint rams from round 0, rams first (`TOP_RAM`, `SPRINT_ANY_ROUND`, `RAM_FIRST`)	40.1%	232	136
+ their ram rule (or ours) with local superiority, sprints from round 0 (`RAM_LOCAL`)	38.9%	238	133
+ keep out of their sprint reach, equals and one longer too (`SPRINT_FEAR 2`)	30.3%	255	86
+ no split while an enemy can reach us (`SPLIT_SAFE`)	35.1%	252	122
+ every worker rushes the centre (`CENTRE_MAX_RUSHERS 99`)	39.4%	239	138
+ teammate spacing at 30% (`SPACING_PCT`)	38.9%	237	122
+ no queen hunt, kill kept / no queen hunt at all	39.8% / 42.3%	227 / 216	122 / 139
without the fight goal / without early trades / without pocket raids	36.2 / 36.5 / 38.2%		

Also tried at 104 games: their hunt pull towards longer enemies (`HUNT_LONGER_PULL`), not fearing longer enemies (`NO_FEAR_LONGER`, `FEAR_MODE`), pocket raids from 2 pearls, a looser portal-pocket room test: all within the noise.

What the rams taught. Their rams decide the game, but neither copying their rule nor avoiding their reach helps us: keeping out of it gave up the food, and their rams barely fell (71 to 61–67 over the same 16 games). What moved was what lets them win the aftermath: our queen's segments working, and the drops.

72 Results

Simulator (13 small maps, seeds 1–16, both sides, 416 games against v-top_team):

	win	eliminated	eliminating
v26.3.0	32.7%	255	109
queen worker + no queen hunt (small maps)	40.4%	222	142
+ no corridor split block	45.0%	203	152
+ portal pocket raids	44.7%	204	151
+ no worker crowd discount	45.7%	205	153
+ both ("c11")	45.4%	206	152
v27.0.0 (+ the portal guard)	**45.7%**	**208**	**154**

Real engine (all 25 maps, seeds 1–4, both sides, 200 games; nothing else running, see below):

- v26.3.0 against v-top_team: 75–125 (37.5%). Small maps 33%, eliminated in 58% of them; large maps 44%.

- c11 against v-top_team: 99–101 (49.5%). Small maps 52.5%: eliminated in 43% of them (was 58%), eliminating it in 45% (was 28%). Large maps 45%. Arena 0 → 62%, Dilemma 0 → 50%, Stripes 0 → 38%; Autarky still 0 of 8.
- c11 against v26.3.0: 122–78 (61%). Small maps 65%, large maps 55%. Queens alive 38 against 55: ours dies on small maps (8 of 60 alive), which cost a few games decided on the queen (Weakhold); Trophy 3–5 and Queen of Spades but She Ages 2–6, mostly by elimination.
- **v27.0.0 against v-top_team: 103–97 (51.5%).** Small maps 51.7%: eliminated in 42% of them (v26.3.0: 58%), eliminating it in 46% (v26.3.0: 28%). Large maps 51% (v26.3.0: 44%). Queens alive 47 against 40. By map, 8 games each: Queen of Spades 8, She Ages 8, Tower Defense 8, Trauma 8, Trophy 8, Australia 7, Maze 6, UNSW 6; lost Autarky 0, Schooltime 0, Islands 1, Weakhold 1, Default Small 2, Devil 2, Small 2, Slithery Fight 2.
- **v27.0.0 against v26.3.0: 115–85 (57.5%).** Small maps 64%: eliminated in 25% of them, eliminating v26.3.0 in 56%. Large maps 47.5% (c11: 55%; the two differ there only in the portal guard, so mostly luck: 8 games a map). Queens alive 38 against 56: on small maps ours is a worker and dies (7 of 60 alive, against 26). Lost: Autarky 0–8, Big Empty 1–7, Help 2–6; won Arena, Dilemma and Tower Defense 8–0.

Running tests alone. Our searches stop at wall-clock deadlines, v-top_team's at node counts, so a busy machine weakens only us. Two earlier real-engine runs made while the simulator shared the two cores gave the same bot (on the large maps the two variants were identical) 40% and 57%. Every real-engine number above was run with nothing else on the machine. The simulator's small maps finish well inside the deadlines and are not affected (v26.3.0: 34% and 35% over the same 104 games, busy and alone).

Not done. Autarky is still lost every time, and their lane economy (relay children farming dead-end lanes) is worth 11 points to them; ours raids pockets only with 3 pearls. The queen on small maps is a worker all game; reverting her at round 200 made no difference against v26.3.0 in the simulator (64% against 66%, 120 games on 15 small maps).

Part XVII
v27.0.0 → v27.1.0: the queen kill on every map

v27.1.0 starts here. Everything before this part is v27.0.0 (section 72).

73 The rule

- **Anyone but our queen rams the enemy queen when it can reach her this turn, on every map.** Workers, growers and the king alike; by a step, or by a sprint that pays segments. It is checked first in the turn, before anything else (`queen.hpp`, `play_queen_kill`). The rammer dies in the head-on whatever it pays on the way, so a long sprint costs nothing a short one does not: one dragon and its length. The only difference is that it leaves fewer drops, and on small maps their swarm eats most of those anyway.
- **Our queen never rams theirs** (dragon 0 or 1), including while she plays as a worker on a small map. v27.0.0 stopped her only as our last unit.
- **The walk towards her stays off on maps under 1,100 tiles.** v27.0.0's switch `KILL_QUEEN_MIN_AREA` turned off both the walk (1,000 a step) and the kill there; it is now `QUEEN_CHASE_MIN_AREA` and turns off only the walk.
- Larger maps: unchanged (anyone but our queen already rammed her there).

Files: `queen.hpp` (`play_queen_kill`, `enemy_queen_seen`, `queen_kill_path` takes its reach) and `main.cpp` (the renamed switch, comments).

Why. v27.0.0 dropped the kill on small maps together with the walk, on a 2.5-point simulator difference inside the noise (section 71). But a reachable queen is at worst a one-for-one trade, and when our queen outlives theirs the kill decides the game at round 500.

74 Results

Simulator.

- Against v-top_team (13 small maps, seeds 1–16, both sides, 416 games): 45% (v27.0.0: 46%); eliminated in 207 games (208), eliminating it in 149 (154).
- Against v26.3.0 (15 small maps, seeds 1–4, 500 rounds, 120 games): 62% (v27.0.0: 64%).
- Also tried: the kill only within our free steps and not by the king or growers: 45% against v-top_team. Same.
- In 10 games against v26.3.0 the kill fired in 6, killing their queen each time. Our queen, a worker on these maps, was usually dead first (rounds 18–92), so mostly it turns "their queen alive" into "both dead", and the longest dragon decides.

Real engine (all 25 maps, seeds 1–4, both sides, 200 games each, nothing else running):

- Against v-top_team: 99.5–100.5 (49.75%; v27.0.0: 103–97). Small maps 50% (51.7%), eliminated in 42% of them (42%); large maps 49% (51%; the two versions play these the same).
- Against v26.3.0: 115–85 (57.5%; v27.0.0: 115–85). Small maps 62.5% (64%), large maps 50% (47.5%). Their queen alive in 52 games (v27.0.0: 56); on small maps 23 of 60 (26).

So against these two the kill neither gains nor costs anything we can measure. It stays on for the reason above: a team whose queen we can reach while ours is alive.

Part XVIII
v27.1.0 → v27.2.0: what we do with what we find

v27.2.0 starts here. Everything before this part is v27.1.0 (section 74).

75 Exploration against v-top_team, measured

The question. v-top_team still eliminates us on most small maps and out-grows us on several large ones. Is it finding food that we do not? Measured in the simulator, both sides, v27.1.0 against v-top_team (4 games a map, 200 rounds; `bed_race2.py`): for each team, the share of the map's pearl-bed spawn rate it has had within 3 tiles of one of its heads by round 100, and the pearls it ate by then; and each bot alone on a map (the other team removed at round 0).

Finding food is not where we lose. Over all 21 maps that last 100 rounds we had seen 83% of the bed rate by round 100 against their 72%, and eaten 156 against 147. We see less only on four maps:

	seen by round 100		eaten by round 100	
map	ours	theirs	ours	theirs
Autarky	61%	100%	48	143
Schooltime	62%	71%	127	186
Weakhold	72%	76%	50	106
Help	85%	89%	186	307
Stripes	100%	100%	42	77
Islands	81%	80%	149	202
UNSW, Maze, Tower Defense, Trauma, both Queens of Spades: we see far more.				

Where they out-eat us, it is what happens on and after reaching the food:

- **Dead-end lanes** (Stripes, Weakhold, Portals). Stripes keeps its food in four lanes of beds that refill every round. Alone there for 100 rounds they walked into lanes 26–32 times and ate 76–91 pearls in them; we did 12–15 and 33–41. Their *lane economy* (section 76) is what we lacked.
- **Weakhold**: a v27.0.0 regression and a queen bug (sections 77, 78).
- **Schooltime**, a maze. Alone, by round 50 they eat 24–39 to our 7–16, on the same coverage. Their dragons have a pearl in view within 6 steps on 34–37% of their turns, ours on 16–25% (when one is, we go for it: 140 of 152 such turns). So their way of choosing where to go when nothing is in view finds food sooner. Not solved here; the bed-frontier rule (section 79) helps. Also both queens are sealed in 2×2 rooms all game (section 78).
- **Help**, open water. Alone they eat 16% more by round 100 (397 against 341); against each other 65% more, and our units fall from 42 at round 50 to 29 at round 100 (8 games) while theirs reach the cap. In one game the head-ons were even (153 of ours died in them, 155 of theirs) but the drops were not. Of 103 head-ons in one game, their nearest other head was closer in 64: they ate 162 of those drops, we ate 11. Where ours was closer (27) we ate 67 to their 21.
- **Autarky**: alone we eat as much as they do (114 against 121 by round 100). We lose the fight for the middle zone in the first 20 rounds, and then the map.

76 The lane economy (maps under 1,100 tiles)

Ported from v-top_team (their own ablation: without it they won 0.43 → 0.55 less against us, section 69). All in `main.cpp` (`build_lanes`, `lane_food`, `play_lane_split`, `lane_run_food`); switches `LANE`, `LANE_AREA 1100`, `LANE_OUT`, `LANE_RELAY`, `LANE_RUN`, `LANE_IN_HOLD_ALL`, `LANE_RELAY_FOOD 2`, `LANE_RUN_MIN 2`.

- **Lanes.** Each turn a dragon peels the map it knows: a seen tile with at most one way on (kelp the other sides; an edge it has not seen, or a portal whose far end it does not know, counts as a way on) is a dead end; take it away and repeat. What comes off are 1-wide trees hanging off the rest by one edge, each tile with its depth.
- **Food in a lane**: the most pearls along one path deeper, counting a pearl seen there in the last 30 rounds and a bed due by the time a head would get there. A tile with another dragon on it (seen in the last 2 rounds) ends the path and marks the lane busy.
- **Walking in** (head deeper than neck): no routine split (the part kept would die at the end with more segments). With nothing left ahead, all but the front 2 split off and walk out (`LANE_OUT`).
- **Walking out with the tail in a lane facing deeper**: with 2 or more pearls behind and nobody in there, split off 2 that walk back in (`LANE_RELAY`); with less, or somebody there, no routine split (its child would face into it).

- **Raids from 2 pearls**: a step deeper into a lane whose food brings us to 4 is a pocket raid (`LANE_RUN_MIN 2`; other pockets keep `POCKET_MIN_PEARLS 3`). A portal pocket's pull counts its beds due within 8 rounds. Never the queen (section 78).

Effect. Stripes alone, 100 rounds: lane entries 16–19 (was 12–15), pearls eaten in lanes 46–52 (33–41), in all 97 (79; theirs 135). Against v-top_team on Stripes (simulator, 32 games): 34% (v27.1.0: 19%; with the lane economy alone added, 41%).

77 Splits in dead ends: the Weakhold regression

What went wrong. v27.0.0 let short dragons split in 1-wide corridors (`CORRIDOR_SPLIT_MAX_LENGTH 0`). On Weakhold, a map of 1-wide pockets of three beds, a worker of 2 walked into a pocket, reached 4 halfway in, split there, and the 2 it kept ate the last pearl and died at the end with 3 segments; whole teams wasted away like this. Simulator, 32 games against v-top_team (500 rounds): v27.1.0 was eliminated in 21 (v26.3.0: 3); real engine, 6 of 8.

The rules (what the corridor block was for, tested directly; `main.cpp`):

- `SPLIT_HOLD_DEAD_END`: when every step our head can take ends in a dead end (a closed region with no loop), a worker splits only if the part it keeps can still split out at the far end (4 counting the pearls on the way). Where the lane economy is on, walking into a lane holds the split instead (`SPLIT_HOLD_WITH_LANES false`).
- `SPLIT_CHILD_ROOM`: no routine split whose child (born at our tail) would start in a trap: the region at our tail closed and too small to turn in, or a dead end (searched 12 tiles past its length).

Weakhold, simulator, 32 games against v-top_team	win	eliminated
v26.3.0	44%	3
v27.1.0	31%	21
v27.1.0 with the corridor block back	41%	3
v27.1.0 + the split rules	38%	5
v27.1.0 + the worker queen raids no pocket, alone	22%	24
v27.2.0	22%	**0**

The eliminations are gone, but not the losses: on Weakhold the side that moves second does better for both bots (v-top_team eats 680–700 as side B against our 490–510), and those games now go to round 500 and are lost on the longest dragon. Real engine: v27.1.0 2 of 8 (6 eliminated), v27.2.0 0 of 8 (1 eliminated); against v27.1.0, 6–2.

78 The queen

- **The worker queen raids no pocket** (`QUEEN_NO_RAID`; `queen.hpp keeps_queen_head`). On maps under 1,100 tiles she plays as a worker (v27.0.0), and that included pocket raids, which by design end with the raider's head dead in the pocket. On Weakhold she did this by round 27 in every game. Now, as for the protected queen: no pocket plan, a trap costs her her life (`trap_cost`), and in a split escape she is the part that lives (she detaches 2 and waits, `QUEEN_WAIT_SPLIT`; `endgame.hpp`).
- **A sealed queen is not crowned** (`SEALED_QUEEN_NO_CROWN`, `QUEEN_SEALED_MAX 16`; `queen.hpp sealed_at`, `crown_ok`). On Schooltime both queens start in a 2×2 room of kelp and end at length 3: the queens tie and the longest dragon decides. Ours was crowned at round 380 all the same, so nobody else became king and the feeders walked to a room they could not enter (real engine: 1 win in 8). Now a queen whose every reachable tile is known and within 16 tiles does not stand, nobody offers her, and the king is elected as if she were not there. Real engine, Schooltime: 2 of 8 (v27.1.0: 1).

79 Larger maps: company before a ram, and the bed frontier

- **On maps of 1,100 tiles or more** (`FIGHT_LOCAL_MIN_AREA`): a worker trades into an enemy head only with at least as many of our heads as theirs within 3 tiles of it (`RAM_LOCAL`), and workers of length 8 or less close on visible enemy heads longer than them, 500 a step (`HUNT_LONGER_PULL`). Both were v27.0.0 tests that did nothing on the small maps. On Help (simulator, 8 games, 200 rounds) our units at round 100 went from 29 to 39 (theirs 64 → 57) and our pearls from 388 to 495 (theirs 779 → 740).
- **The bed frontier** (`BED_FRONTIER_BONUS 4`, every map): a never-seen tile next to a bed we know is an exploration objective at cost 4 instead of 8. Beds come in clusters, so the swarm explores round the ones it finds.

Simulator, 10 large maps, both sides, 500 rounds, against v-top_team: without the bed frontier 42% and 38% (seeds 1–4 and 5–8, 80 games each); with it 54% and 46%. On the real engine the large maps did not move (below).

80 Tried, and left off

Simulator, against v-top_team. All are switches in `Config`, off unless noted.

test	result
the worker queen starts no head-on (`QUEEN_WORKER_NO_TRADE`)	Weakhold 34% (38%)
the queen protected again from round 380 on small maps (`QUEEN_SMALL_UNTIL 380`)	Weakhold 34% (38%)
halving splits while the team is under half the unit limit, as theirs (`SPLIT_HALF`)	small maps 42% (41%)
lane raids from 3 pearls (`LANE_RUN_MIN 3`), relays from 3	Weakhold 31–38% (25%), Stripes 28% (41%)
walking into a lane, split if the kept part can still split out (`LANE_IN_HOLD_ALL false`)	Prisoners Dilemma still lost
the endgame 50 rounds earlier (gather 330, sacrifice 350)	large maps 29% (42%), 15 eliminations (2)
on large maps: not fearing equal or longer heads (`FEAR_MODE 2`); their ram rule (`TOP_RAM 2`)	Help: little or no change in units or food
bed frontier at 6	large maps 47% (bed frontier 4: 46%)
scouts (`SCOUT_PERMIL 333`); rich spots over sonar (`RICH_SHARE`, mode 1)	small maps 39% and 41% (42%)

81 Results

Simulator (against v-top_team):

- 13 small maps, seeds 1–16, both sides, 416 games: 44% (v27.1.0: 45%). Stripes 34% (19%), Default 47% (28%), Tower Defense 94% (84%); Prisoners Dilemma 0% (50%). Prisoners Dilemma is the same game on every seed (the simulator and the real engine play it move for move alike), so it counts as 2 games, not 32; any change to how the first long dragon splits in its pocket turns our side-B win into a loss.
- Weakhold (32 games, 500 rounds): eliminated in 0 (21).
- 10 large maps (80 games, 500 rounds; two seed sets): 54% and 46% (v27.2.0 without the bed frontier: 42% and 38%).

Real engine (all 25 maps, seeds 1–4, both sides, 200 games each, nothing else running):

- **Against v27.1.0: 111–89 (55.5%).** Small maps 59% (Autarky 8–0, Weakhold 6–2, Portals 6–2, Devil 6–2), large maps 50%. Queens alive 50 against 43.
- **Against v26.3.0: 116–84** (v27.1.0: 115–85). Small maps 65%, large maps 47.5%.
- **Against v-top_team: 92.5–107.5** (v27.1.0: 99.5–100.5). Small maps 46% (50%), large maps 46% (49%). Prisoners Dilemma 0 of 8 (4 of 8: the side-B game above, four times); Default 5 (2), Schooltime 2 (1), Help 3 (1); Weakhold 0 (2), Trauma 6 (8). Autarky is still 0 of 8.

So v27.2.0 beats v27.1.0 head to head and keeps its margin over v26.3.0. Against v-top_team it is 7 games lower over 200; 4 of those are Prisoners Dilemma and the rest is within the luck of 200 games (±7). The lane economy and the dead-end split rules show their gain against v27.1.0 rather than against the bot they were taken from.

Not done. Against v-top_team: Autarky (the middle zone in the first 20 rounds), where their dragons go when no food is in view (Schooltime), and the drops after head-ons on open maps (Help, Islands).

Part XIX
v27.2.0 → v31.0.0: Leo's v30 as the base, the queen first; v40.0.0: one bot from six

82 Why a new base

Leo's v30.0.0 plays move for move like v-top_team. Against it v27.2.0 scores 46.6% (116.5–133.5 over 250 games, both sides, on the test rig below); the user's own 250 games with v27.2.0 always as team A gave 112–137. v31.0.0 therefore starts from v30's code: `bots/New/current/bot-v31.0.0` is a copy of `bot-v30.0.0-leo` (it builds to the same program and plays the same games), and v31.0.0 changes two switches and one line of it.

Where v30 loses to v27.2.0 is the queen. With v27.2.0 as team A, v30's queen was the shorter one at round 500 in 14 of 250 games, v27.2.0's in 4: v27.2.0 crowns its queen on big maps and feeds her, v30's queen is never fed (on big maps she stays within 8 tiles of home and takes one step a turn).

83 What v31.0.0 changes

- **The queen is the king from round 250** (`QKING_FROM 250` in `tuning.hpp`; Leo's switch, 0 in v30). She beacons as the king; from the guard round the short dragons stay round her, and from the feed round they give her their bodies. A dragon that has not seen or heard her for 30 rounds (`QKING_SILENT`) goes back to the longest dragon as its king.
- **On big maps the crowned queen free-sprints** (`Q_KING_SPRINT`; `decide.hpp`, the free sprint). In v30 every dragon but the big-map queen takes the free steps its length gives it towards food in view; she always took one. From the round she is crowned she sprints like the others. Her escape sprint, her leash to home and her other rules stay as in v30.
- `main.cpp`: the header comment names the version.

Files: `tuning.hpp`, `decide.hpp`, `main.cpp`.

84 The test rig

The games now run on the user's Mac, in the Cowork Linux virtual machine (4 cores), in `random/claude-tests/`: the official engine built from source (unswbc 1.2.7; a stand-in for `std::format`, which g++ 11 lacks), both bots compiled from the project folder, and `ubctest.py`, which plays every (map, seed) from both sides, runs as many games at once as memory allows (a 64×64 game needs about 1.2 GB: every dragon is a process) and resumes across the 3-minute calls. 250 games against v27.2.0 take about 6 minutes, against v30 about 2.

v27.2.0 plays the tests with its deadlines off (`BOT_UNLIMITED`). On that machine its turns use under a tenth of the budget on 97% of turns and a deadline cut 0.002% of them; with deadlines on, 9 of 250 games came out differently on a second run, with them off none do. Every game is then repeatable, so a variant and the base are compared game by game.

85 Results

Against v27.2.0: all 25 maps, seeds 1–10, both sides, 500 games each. "Lost by" counts the variant's losses; the last column, the games whose result changed against v30.

			lost by			changed
variant	score	elim.	q. dead	q. shorter	longest	total	+ / −
v30	280.5 (56.1%)	107	23	22	66	1	
the queen king from 250	281.5 (56.3%)	107	23	23	65	0	19 / 18
big-map queen sprints, all game	294.5 (58.9%)	109	21	9	64	2	45 / 31
only with no enemy in view	283.5 (56.7%)	107	25	16	66	2	31 / 28
king from 300, sprints as king	290.5 (58.1%)	107	23	9	70	0	22 / 12
v31.0.0: king from 250, sprints as king	**298.5 (59.7%)**	107	24	6	64	0	27 / 9

Against v30 (that is, v-top_team's play): seeds 1–10, both sides, 500 games each. v30 against itself scores exactly 50%: each (map, seed) is the same game from both sides.

		lost by (ours / theirs)	
variant	score	queen dead	queen shorter
the queen king from 250	268–232 (53.6%)	22 / 41	11 / 18
big-map queen sprints, all game	259–241 (51.8%)	30 / 24	2 / 24
only with no enemy in view	259–241 (51.8%)	22 / 22	5 / 25
king from 250, sprints all game	260–240 (52.0%)	22 / 19	2 / 29
king from 300, sprints as king	265–235 (53.0%)	27 / 33	3 / 26
v31.0.0	**265–235 (53.0%)**	27 / 27	1 / 28

The queen king from 300 alone (no sprints): 392–358 (52.3%) over 750 games.

The crowning round alone against v27.2.0 (seeds 1–5, 250 games; v30: 134.5): 200: 138.5, 250: 139.5, 300: 138.5, 350: 134.5, 400: 132.5, 430: 135.5, 450: 133.5, 470: 133.5. From 430 on almost no game changes (one at 470): the feeding has too little time left.

In the user's format (v27.2.0 as team A, the v31 bot as team B; 250 games, seeds 1–10):

	v30	v31.0.0
final A – B	114 – 136	101 – 149
lost by elimination (A / B)	72 / 53	72 / 53
lost by queen (its queen dead)	12 / 9	9 / 11
lost by queen (its queen shorter)	4 / 14	11 / 3
lost by longest dragon	45 / 37	54 / 34
lost by total length	3 / 1	3 / 0

What it shows.

- Crowning alone hardly changes the games against v27.2.0 (which crowns its own queen) but beats v30, whose queen is never fed.
- Sprints are what make the crowned queen long. Sprinting all game, she died more often against v30 (30 against 24); sprinting only once crowned, her deaths stay level (27 and 27) while her losses on a shorter queen against v27.2.0 fall from 22 to 6 of 500 (Trauma 12 → 5, Stronghold 6 → 0, Maze 3 → 1).
- Eliminations do not move in any variant: these rules act late, on big maps.

86 Tried, and left off

The crowning rounds 200, 300, 350, 400, 430, 450 and 470; big-map queen sprints all game, or only with no enemy in view; crowning from 250 with sprints all game (tables above). On v27.2.0, not shipped: the worker queen taking no fight goal (+1 in 250 games against v30) and the last dragon never starting a head-on (+0.5).

Not done. v31.0.0 against other opponents (the panel in `battlecode-automation/panel.json`, older versions) and a `-sandbox` check; the queen's deaths (24 of 500 against v27.2.0: Maze 9, UNSW 6).

Part XX
v31.0.0 → v40.0.0: one bot from six

87 The brief, and the bots

Six bots were on the table: our v31.0.0 (Leo's v30.0.0 with the crowned queen), Leo's v30.6.0, the top team's v3.9f, our old line's v27.2.0, and two Codehammer-derived bots from the ladder, noobergamer (v73, rank 39) and cybers (v68, rank 26). v40.0.0 was to take what each does better and beat all of them, with one rule from the user: no map recognition, no per-map strategy and no map knowledge seeded at round 0 (the qualifier is played on new maps). Everything v40 does follows from what its dragons see.

A round robin of the six on the Mac runner (all 25 maps, seeds 1–6, both sides, 300 games a pair, 4,500 games) set the baseline: v30.6.0 56.6%, v31.0.0 56.2%, top team 55.0%, v27.2.0 46.3%, noobergamer 40.1%, cybers 39.7%. The three top-team-family bots are even with each other (v31–v30.6: 49.3/50.7). Where they differ is by map: v31 loses Tower Defense (17%), Trophy (23%), Queen of Spades (33%), Australia (27%) and Maze (35%); v30.6 loses Devil (38%) and Australia (43%). The Codehammer family wins exactly the maps the top-team family loses: v27.2.0 scores 94% on Queen of Spades, 88% on Tower Defense and 56% on Trophy against the field.

88 Why the top-team family loses Tower Defense, Queen of Spades and Trophy

Every one of those losses is an elimination, and the death logs show two steps.

The start is barren and the food is elsewhere. On Tower Defense the dragons start in towers of dead tiles rimmed with slow beds (gaps of up to 1,000 rounds); the food is in the central structure, 15–22 steps away through a one-wide exit. On Queen of Spades the start opens onto a corridor of 1–1,000-gap beds; the fast beds (gap 1–50) sit in two walled pens. On Trophy the fast beds are in the cup. v30's navigation prices a known pearl at its arrival time and unseen water at depth +3, so a pearl due in 5–12 rounds at home always beats the exit 20 steps away. A dragon in the tower takes one trickle pearl, replans, and takes the next; the whole team stays home. v31

eats 3–9 pearls by round 100 on Tower Defense; v27.2.0, whose workers disperse to unexplored 7×7 blocks from round 0 and share rich spots by sonar, eats 80.

Then the swarm trades heads. v27.2.0 values one of its units at 2 segments and one of ours at 3, so it rams any enemy no longer than itself from its third unit on (noobergamer doubles the aggression). v30's threat model assumes an enemy rams only when it gains length, so an equal-length enemy is no threat at all. Against two to five dragons that never grew, a few even trades are an elimination.

89 What v40.0.0 is

`bots/New/current/bot-v40.0.0` starts from `bot-v30.6.0-leo` (Leo's right of way, the top team's opening and queen rules) and adds, every switch in `tuning.hpp` (a `-DV40_X=value` build sets one for a test):

- **v31's crown on every map** (`QKING_FROM 250`, `QKING_AREA 0`, `QKING_SILENT 30`, `Q_KING_SPRINT`): the queen is the king from round 250 and the crowned big-map queen free-sprints.
- **FORAGE** (`navigate.hpp`): poor water is fewer than 4 pearls due within 40 rounds per 100 known tiles in the 17×17 square round the head (maps under 1,100 tiles, before round 380). In poor water unseen water is worth 10 steps of the way (its cost is depth $+3-10$ instead of depth $+3$, half of that for water not looked at for 70 rounds) and the route search looks 10 steps deeper before it settles for the cheapest goal. A known pearl keeps its price (its arrival time), so the dragon still takes a pearl due in 3 rounds next door, but no longer walks 12 steps for one due in 12 while unseen water lies 15 steps away; and a route to unseen water is kept against a new goal by 6 instead of 2 (`FORAGE_STICK`). Food in view is eaten as before.
- **QHUNT** (the top team's late queen hunt, `decide.hpp`, `perceive.hpp`): from round 250 a dragon that sees their queen's head beacons her tile (a kind-1 beacon with id 1, relayed like the long-enemy beacon); every dragon keeps her last known tile for 60 rounds, drops it when the spot is in view and empty, and dragons of at most 8 segments go for it while our king is at least as long as the longest enemy seen lately, so that her death wins the tiebreak. v30 already rammed her at any price once in view.

90 The test rig

The Mac runner (`random/claude-tests/mac/runner.py`): a queue of build and match jobs played by the official engine on all 14 cores, bots compiled with clang (libc++, which is what the judge compiles to wasm; v27.2.0-line bots play different games under GCC's libstdc++: 36 of 100 identical), 14 games at once, 2.25 games a second. Candidates are compared with v31 game by game on the same seeds (`panel.py sum NAME -base v31`); all opponents are deterministic but noobergamer and cybers, which think by the clock and rarely differ.

91 Results

Candidates (all 25 maps, seeds 1–4, both sides, 200 games a pair, clang builds). "Five" is the mean score against the five bots other than v31; the losses are counted over those 1,000 games; "+ / –" counts the games won / lost that v31 lost / won on the same (map, seed, side). v31 itself against the five: 54.6%.

	score against								lost by		
variant	v31	v306	top39f	v272u	noober	cybers	five	elim.	q. dead	q. shorter	l
v31	–	51	50	54	59	59	**54.6**	214	67	12	
v40 base (v30.6 + crown)	51	52	51	53	64	64	**56.6**	200	66	11	
+ FORAGE	57	54	54	55	63	62	**57.5**	187	71	10	
+ FORAGE + EVEN 3	46	46	50	54	58	60	**53.4**	231	60	12	
+ FORAGE scouts + EVEN, teams of at most 6	48	47	49	56	62	63	**55.4**	214	68	9	
+ FORAGE scouts, trickle pearls no goal	50	50	54	55	63	63	**56.8**	204	72	9	
+ FORAGE, right of way off	54	50	52	56	67	65	**58.2**	192	71	9	
+ FORAGE + QHUNT (= v40.0.0)	55	52	54	55	64	64	**57.6**	187	70	9	

- **the base** (v30.6 + the crown everywhere + sprint) gains 2 points on v31: it inherits v30.6's wins on Tower Defense, Trophy and Australia and v31's late queen; it also inherits v30.6's Devil (78 → 46%).
- **FORAGE** adds another point and moves the four target maps: Tower Defense 15 → 44%, Trophy 22 → 42, Queen of Spades 32 → 46, Queen of Spades but she ages 32 → 54 (and Default 38 → 52, Islands 40 → 56). Against the top-team family those maps are now won (Tower Defense 7–1 against v31, Queen of Spades but she ages 8–0 against the top team); against the Codehammer family they are still lost by elimination (0–8 against v27.2.0 on Queen of Spades): the swarm's head-on trades remain.
- **EVEN**, the cure for the trades, costs more than it saves everywhere: 4 points against the top-team family, and even 2–3 against noobergamer and cybers. Keeping out of an equal dragon's reach means keeping away from contested pearls, and on these maps the pearls are the game. Restricted to teams of at most 6 dragons (`EVEN_UNITS`) it still costs 3 points against the top-team family: the opening is where those games are decided. It stays in the code, off.
- **Scout destinations** (half the dragons head for unseen 7×7 blocks from a sequence keyed by their id, and a trickle pearl is no goal in poor water, as v27.2.0 does) won Queen of Spades and Trophy against v27.2.0 in single simulator games but lost a point on the panel, 7 against v31 and 4 against v30.6: dragons walking to far blocks do not eat. The simple bonus ships; the destinations stay in the code, off.
- **Right of way off** gains 3 against noobergamer and cybers and loses 4 against v30.6 and 2 against the top team; Devil only comes back to 54%. Leo's rule stays on.
- **QHUNT** is neutral on the panel (57.6 against 57.5, the same losses by every condition) and kills their queen in 5.4% of games instead of 3.9%. It ships on.

Validation on fresh seeds: the shipped build (no flags) against all six bots on seeds 7–12 (unused in any tuning), 300 games a pair, 1,800 games; v31 on the same seeds against the five for comparison.

	score against								lost by		
variant	v31	v306	top39f	v272u	noober	cybers	five	elim.	q. dead	q. shorter	l
v31	–	50	52	61	62	61	**57.1**	303	78	15	
v40.0.0	55	54	55	57	65	66	**59.4**	298	69	15	

By map (seeds 7–12, both bots against the five other opponents, 60 games a map each):

map	v31	v40.0.0	diff
Colosseum	65	65	+0
arena	68	70	+2
australia	37	33	-3
autarky	72	72	+0
big_empty	53	60	+7
default	70	63	-7
default_small	72	75	+3
devil	53	50	-3
dilemma	73	80	+7
help	62	72	+10
islands	53	53	+0
maze	52	52	+0
portals	62	62	+0
queen_of_spades	33	52	+18
queen_of_spades_but_she_ages	20	30	+10
schooltime	45	45	+0
slithery_fight	60	68	+8
small	77	77	+0
stripes	68	73	+5
stronghold	83	88	+5
tower_defense	20	32	+12
trauma	72	67	-5
trophy	55	45	-10
unsw	48	48	+0
weakhold	53	53	+0

What it shows.

- On seeds it was never tuned on, v40.0.0 scores 59.4% against the five (v31: 57.1) and 55% against v31 itself; game by game it turns 212 of v31's losses into wins and 177 of its wins into losses. Every loss category is equal or better: eliminations 298 against 303, queen dead 69 against 78, queen shorter 15 against 15, longest dragon 222 against 244.
- The gains are against the top-team family (+4, +3), noobergamer (+3) and cybers (+5); against v27.2.0 v40 is 4 points worse on these seeds and 1 better on seeds 1–4: our old line is the one opponent it does not beat more than v31 does. Its specialities, Queen of Spades (0–8 against it on seeds 1–4) and Tower Defense, are still its.
- By map the four targets move the most (Queen of Spades +18, Tower Defense +12, Queen of Spades but she ages +10); Trophy is −10 on these seeds after +19 on seeds 1–4, so call it even; 60 games a map is ±13 points.

92 Tried, and left off

EVEN (always, and only while the team has at most 6 dragons), the scout destinations with trickle pearls excluded (tables above); SWARM (an outnumbered dragon takes every enemy for a rammer; queued and cancelled for time, so not measured); the poor-water threshold at 8 instead of 4 pearls per 100 tiles (simulator: Default's economy collapses, 16 pearls by round 100 against v31's 58); `FORAGE_W` 15 (54 / 51 / 52 against v31 / v30.6 / the top team, W 10: 57 / 54 / 54). Not done: the crown without the sprint on small maps (cancelled for time); a `-sandbox` run of the final bot (the user's Mac has the toolkit); Tower Defense itself, where v40 still loses to the Codehammer family (the towers' one-wide exits, friends in the way) and Devil, where v30.6's economy stalls with or without right of way.